

SHANDON J. HERFT

D E S I G N P O R T F O L I O

Selected Works 2021-2026 | CORE studio

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Education

Carnegie Mellon University August 2023-May 2025

Master of Science in Computational Design

Illinois Institute of Technology August 2018-May 2023

Bachelor of Architecture

Specialization in Digital Design and Architecture History and Theory

Master of Engineering in Construction Engineering and Management

Work Experience

Core studio at Thornton Tomasetti July 2025-Current

New York, New York

Full-Time Computational Designer

Kohn Pedersen Fox Associates PC June 2024-August 2024

New York, New York

Full-Time Computational Design Intern

Carnegie Mellon University August 2023-May 2025

Pittsburgh, Pennsylvania

Research Assistant: Architectural Robotics
Teaching Assistant: 48-619 Machine Intelligence, Cybernetics, and Design
Teaching Assistant: 62-715 Shaping Environments
Research Assistant: Rethinking Automation in Construction
Part-Time SAIL Platform Course Development Intern

Chicago Transit Authority January 2023-April 2023

Chicago, Illinois

Part-Time Engineering and Architectural Intern

PALMA August 2022-December 2022

Chicago, Illinois

Part-Time Architectural Intern

Gensler June 2022-August 2022

Chicago, Illinois

Full-Time Architectural Intern

Key Concept Design June 2021-July 2021

Chicago, Illinois

Full-Time Architectural Intern

Skills

Programming:

Python, Java, JavaScript, Kotlin, Arduino, C, C#, HTML/CSS, Bootstrap, Git, RAPID, R, Bash, OpenCV

Drafting/Modelling

Revit, Dynamo, SketchUp, Rhino3D, Grasshopper/2.0, Maya, Blender, AutoCAD, Houdini, OpenStudio, ShapeDiver, 3D Printing, Laser Cutting

Presentation

Photoshop, Illustrator, InDesign, Figma, Sketch, WordPress, Microsoft Office, Miro, Google Suite

3D Visualization/Video Editing

Enscape, 3ds Max, V-Ray, Unity, Unreal Engine, Houdini, Lumion, Twinmotion, Premiere Pro, Cinema 4D

Construction

RS Means Data, Bluebeam Revu, ETABS, Primavera P6, Tekla, SAP2000, Civil 3D, Bentley Infrastructures, RAM, Solibri, 360 Glue

References

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Director CORE Modeling

Madeleine Eggers KPF, New York
mbe42@cornell.edu

Former Senior Computational Designer

Daragh Byrne Carnegie Mellon University, Pittsburgh
daraghbb@andrew.cmu.edu

Track Chair, Computational Design

Alphonso Peluso Illinois Institute of Technology, Chicago
peluso@iit.edu

Studio Associate Professor

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JointLab

Computational Design Thesis

Year: *Fall 2023 - Spring 2025*

Type: *Academic Work*

Advisors: *Joshua Bard, Daragh Byrne,
Vernelle A. A. Noel*

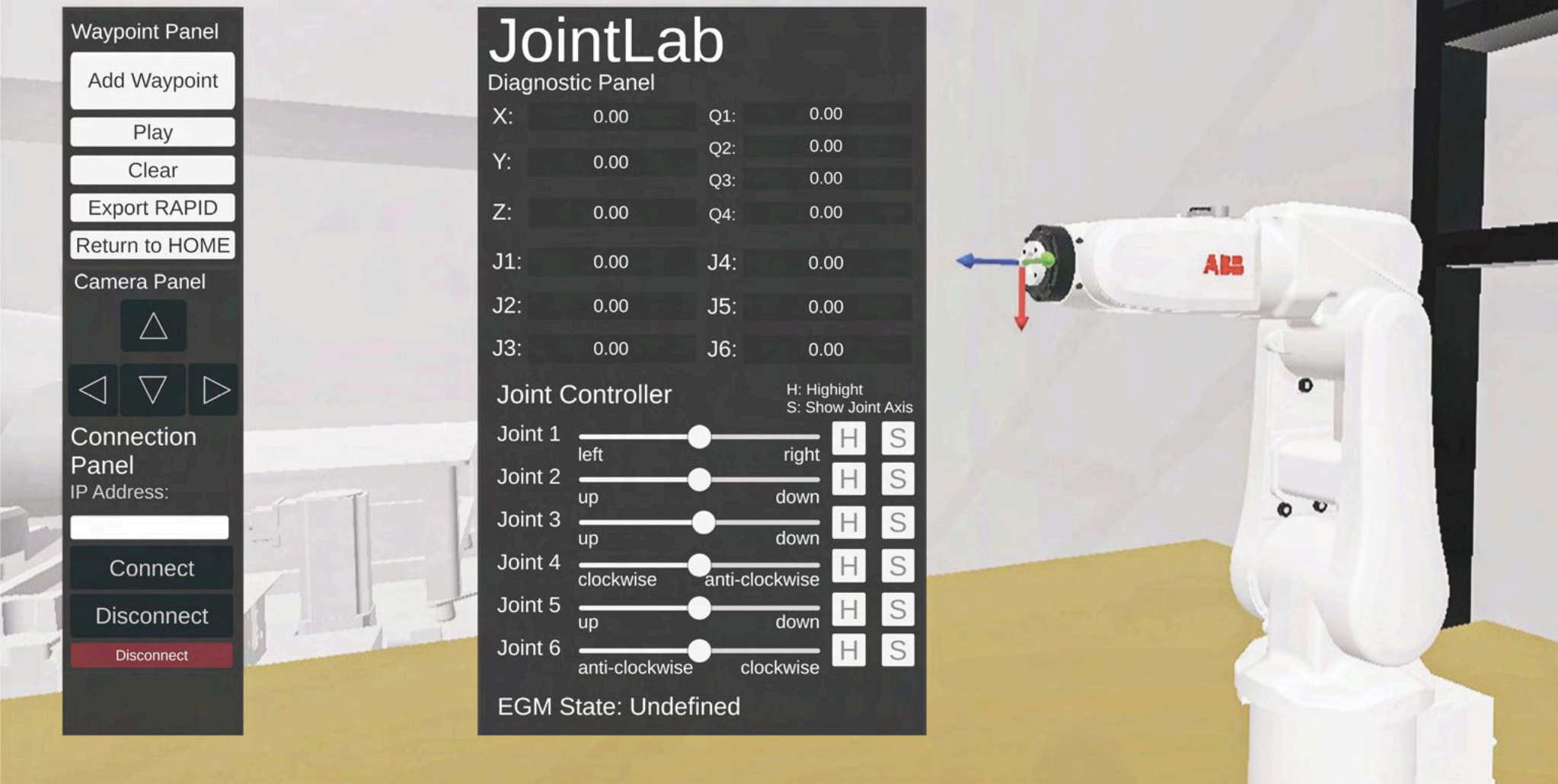
JointLab addresses the critical accessibility gap in architectural robotics education where the high cost of equipment and significant safety risks often restrict direct practical learning. Traditional teaching models create a steep learning curve that separates digital simulation from physical operation which leaves students unprepared for hardware complexities. To resolve this I developed a comprehensive Virtual Reality framework that functions as a precise digital twin for the ABB IRB 120 robotic arm. This project serves as a pedagogical bridge that allows students to build essential kinematic intuition and spatial understanding in a completely safe environment before they ever step into the physical lab.

The system visualized in the interface screenshots on this page is built on a custom Unity application that communicates directly with the robot controller. By integrating Externally Guided Motion and Robot Web Services the framework ensures that the virtual robot mirrors the exact constraints and physics of the actual hardware used in the CMU dFab Lab. The user interface shown here was specifically designed to visualize complex robotic data like axis limits and singularities in an intuitive and gamified format. This allows users to manipulate the robot tool utilizing standard VR controllers and transforms abstract programming tasks into embodied physical interactions.

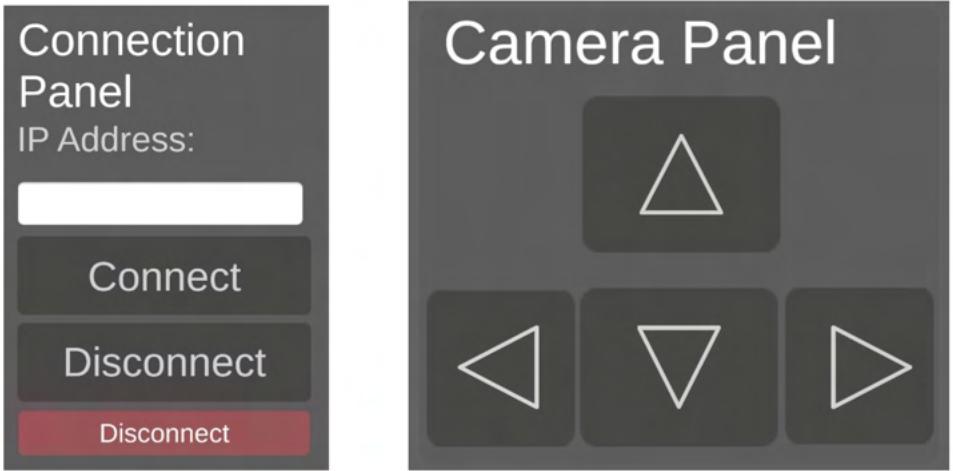
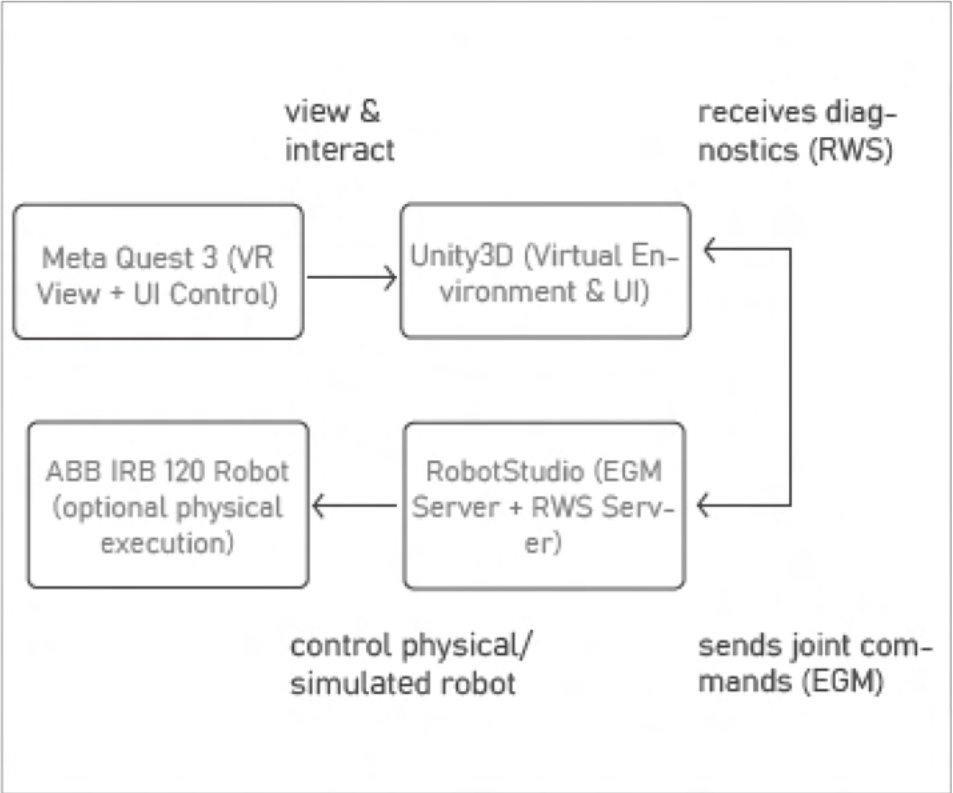
Beyond simple simulation the workflow demonstrated here generates valid RAPID code which validates the tool as a practical manufacturing resource. As the user study data indicates participants with no prior robotics experience were able to successfully plan complex paths and execute tasks solely through this immersive environment. By providing immediate visual feedback on potential errors JointLab effectively lowers the entry barrier for computational fabrication and shifts the focus from technical troubleshooting to creative design exploration.



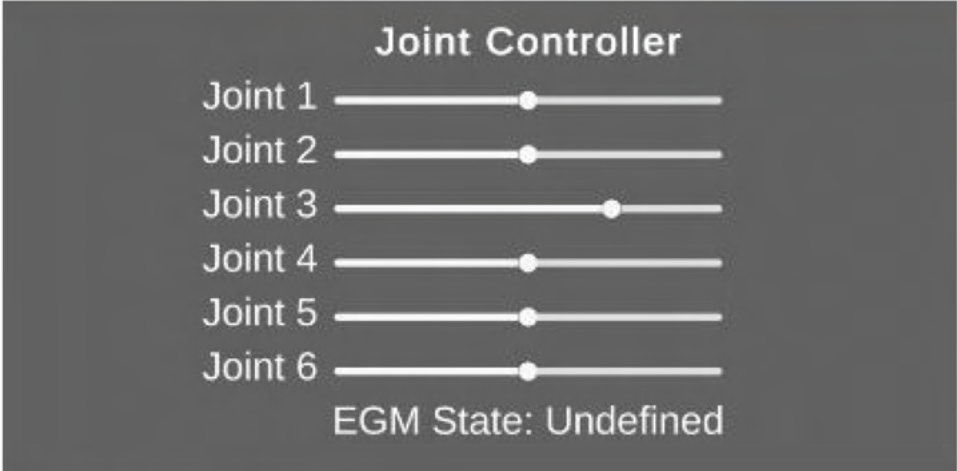
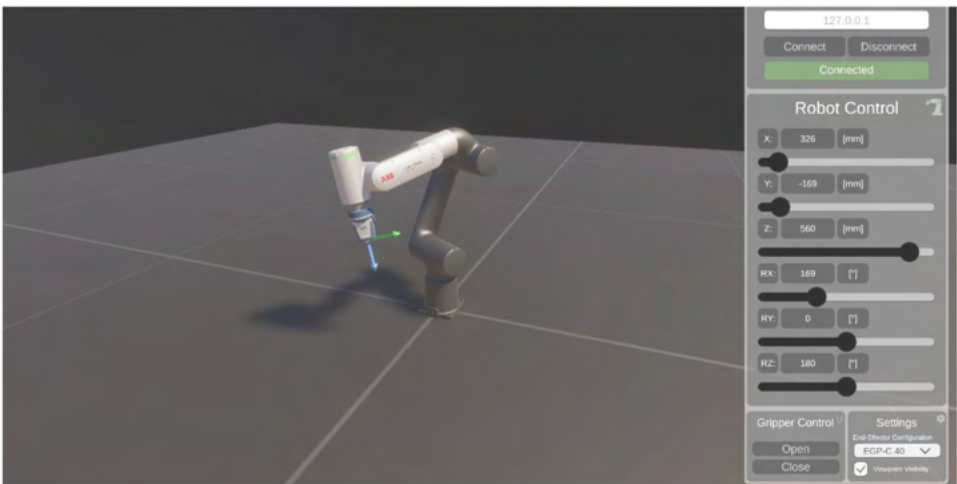
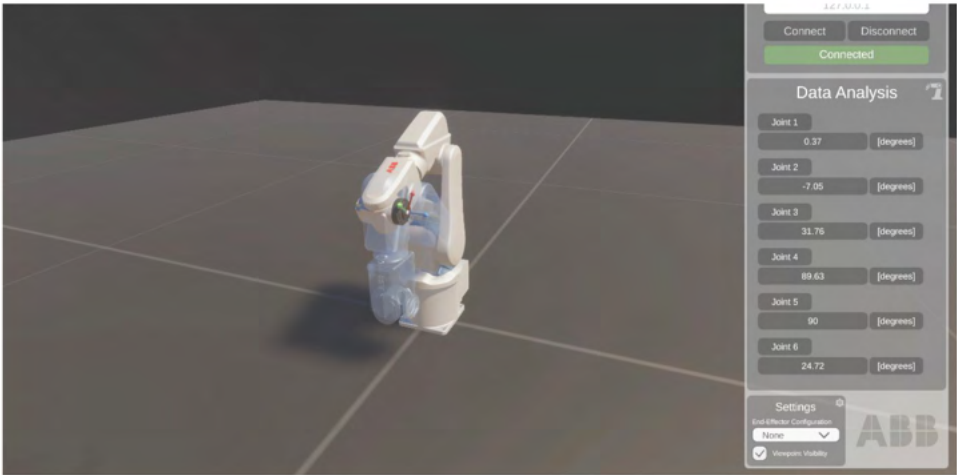
Scan Me



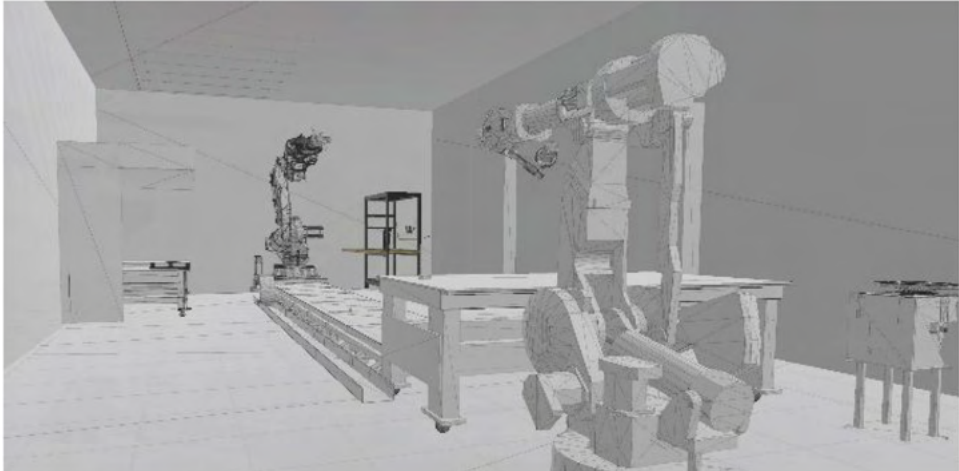
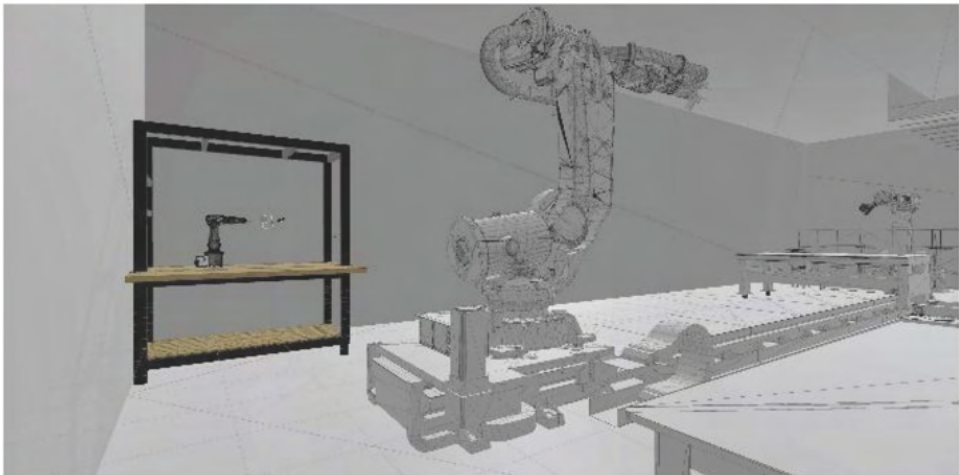
The technical foundation of JointLab draws directly from the precedents shown in the four examples on this page, specifically the digital twin experiments conducted by Roman Parak. His work validated the use of Robot Web Services and Externally Guided Motion for linking Unity with ABB hardware, yet these tests remained limited to 2D monitoring. My research expands this architecture into a fully immersive VR framework that shifts the focus from passive data observation to active embodied control. To ground this interaction in reality I developed a high-fidelity digital recreation of CMU's dFab Lab which serves as the central simulation environment. This digital twin utilizes precise CAD data to ensure that every virtual constraint perfectly mirrors the physical lab space. The user interface manages this bridge through the Connection Panel which executes the necessary IP handshake with the robot controller, while the Camera Panel enables users to navigate the virtual space and verify toolpaths from multiple safety vantage points before physical execution.



Connection Panel and Camera Panel



Prior Critical Investigations

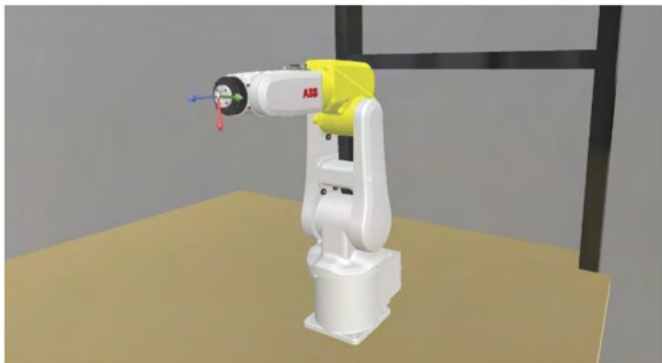


Key Case Studies

JointLab

Diagnostic Panel

X:	0.00	Q1:	0.00
Y:	0.00	Q2:	0.00
Z:	0.00	Q3:	0.00
J1:	0.00	J4:	0.00
J2:	0.00	J5:	0.00
J3:	0.00	J6:	0.00



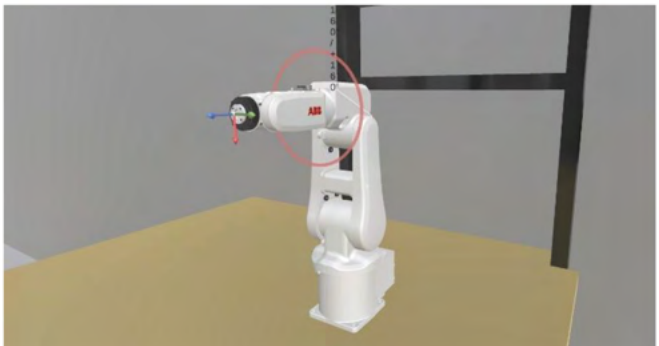
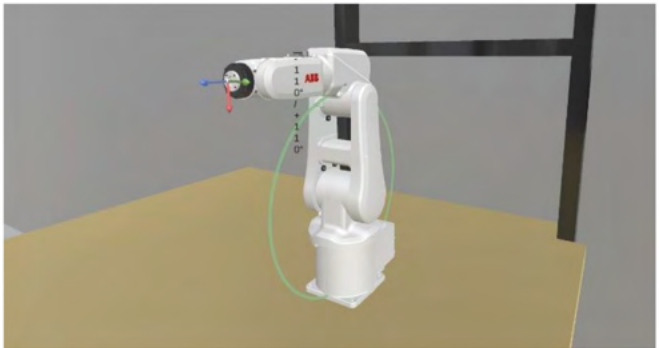
Joint Highlighting Function

Joint Controller

H: Highlight
S: Show Joint Axis

Joint 1	<input type="range"/>	<input type="range"/>	H	S
	left	right		
Joint 2	<input type="range"/>	<input type="range"/>	H	S
	up	down		
Joint 3	<input type="range"/>	<input type="range"/>	H	S
	up	down		
Joint 4	<input type="range"/>	<input type="range"/>	H	S
	clockwise	anti-clockwise		
Joint 5	<input type="range"/>	<input type="range"/>	H	S
	up	down		
Joint 6	<input type="range"/>	<input type="range"/>	H	S
	anti-clockwise	clockwise		

EGM State: Undefined



Joint Axis Visualization

Waypoint Panel

Add Waypoint

Play

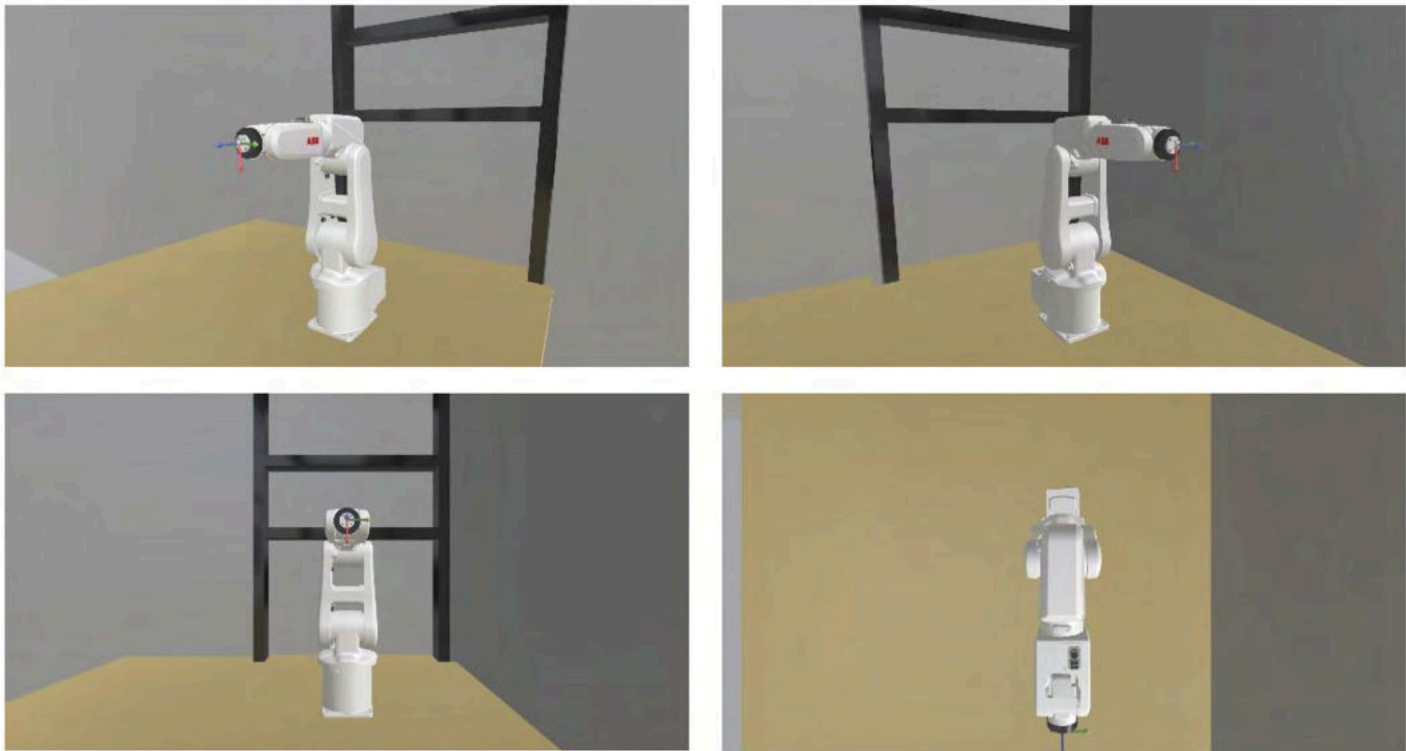
Clear

Export RAPID

Return to HOME

Waypoint Panel

The interface is anchored by the Joint Controller which facilitates direct manipulation of the robot through dedicated sliders for each movement axis. To aid spatial understanding I implemented visual cues such as dynamic movement rings and joint highlighting which instantaneously identify the active axis of rotation and its specific limits on the digital model. This system works in tandem with the Diagnostic Panel a real-time dashboard that displays specific joint angles and Cartesian coordinates. This dual-feedback mechanism ensures that students can visually correlate their manual inputs with precise kinematic data to prevent errors before execution.



Different Camera Views



Different Camera Views

Waypoint Panel

Add Waypoint

Play

Clear

Export RAPID

Return to HOME

Camera Panel

4

Connection Panel

IP Address:

Connect

Disconnect

Shandon Herft CMU MSCD 24

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JointLab

Diagnostic Panel

X: 0.00

Q1: 0.00

Y: 0.00

Q2: 0.00

Z: 0.00

Q3: 0.00

J1: 0.00

Q4: 0.00

J2: 0.00

Q5: 0.00

J3: 0.00

Q6: 0.00

Joint C

Joint 1

Joint 2

Joint 3

Joint 4

Joint 5

Joint 6

left

up

up

clockwise

up

anti-clockwise

right

down

down

anti-clockwise

down

clockwise

H

S

H

S

H

S

RT: Highlight

S: Show Joint Axis

EGM State: Undefined

Task 1

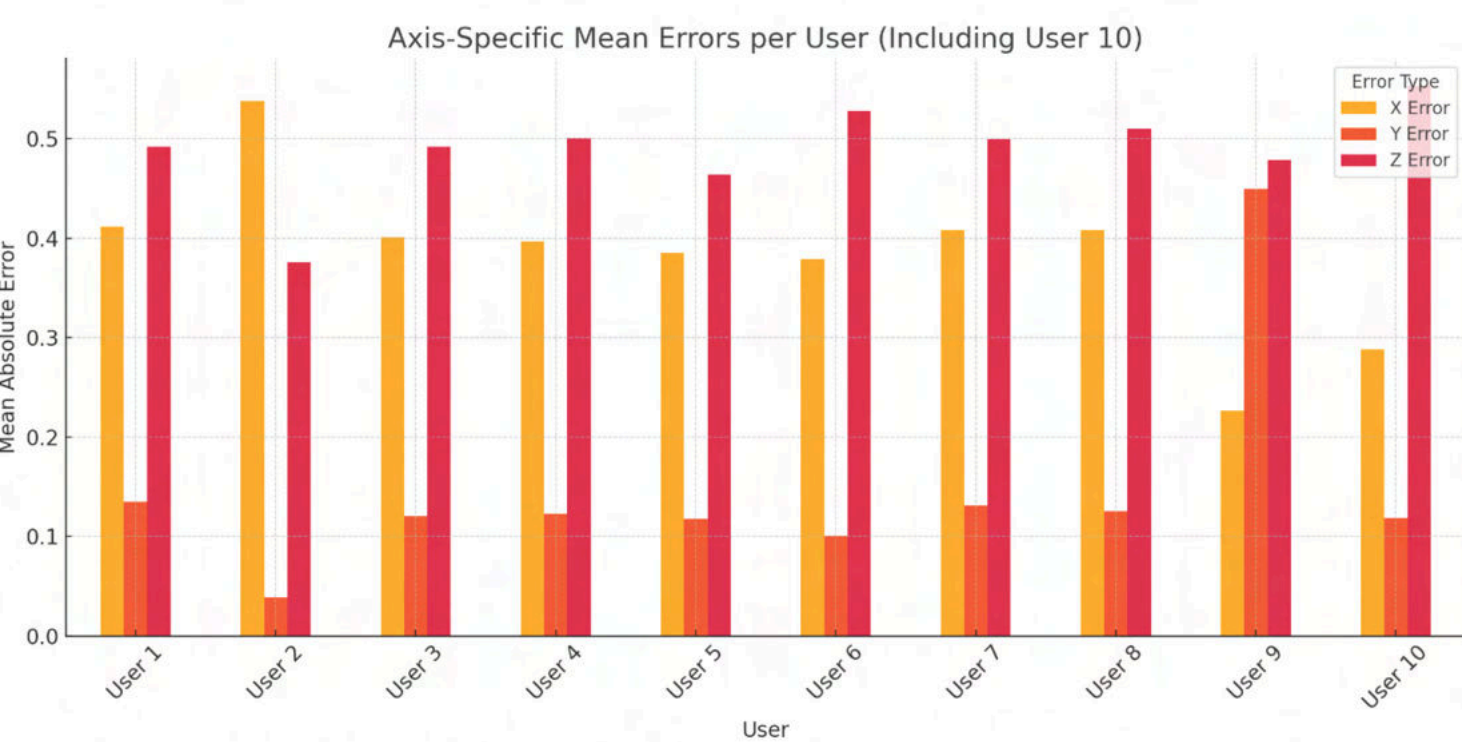
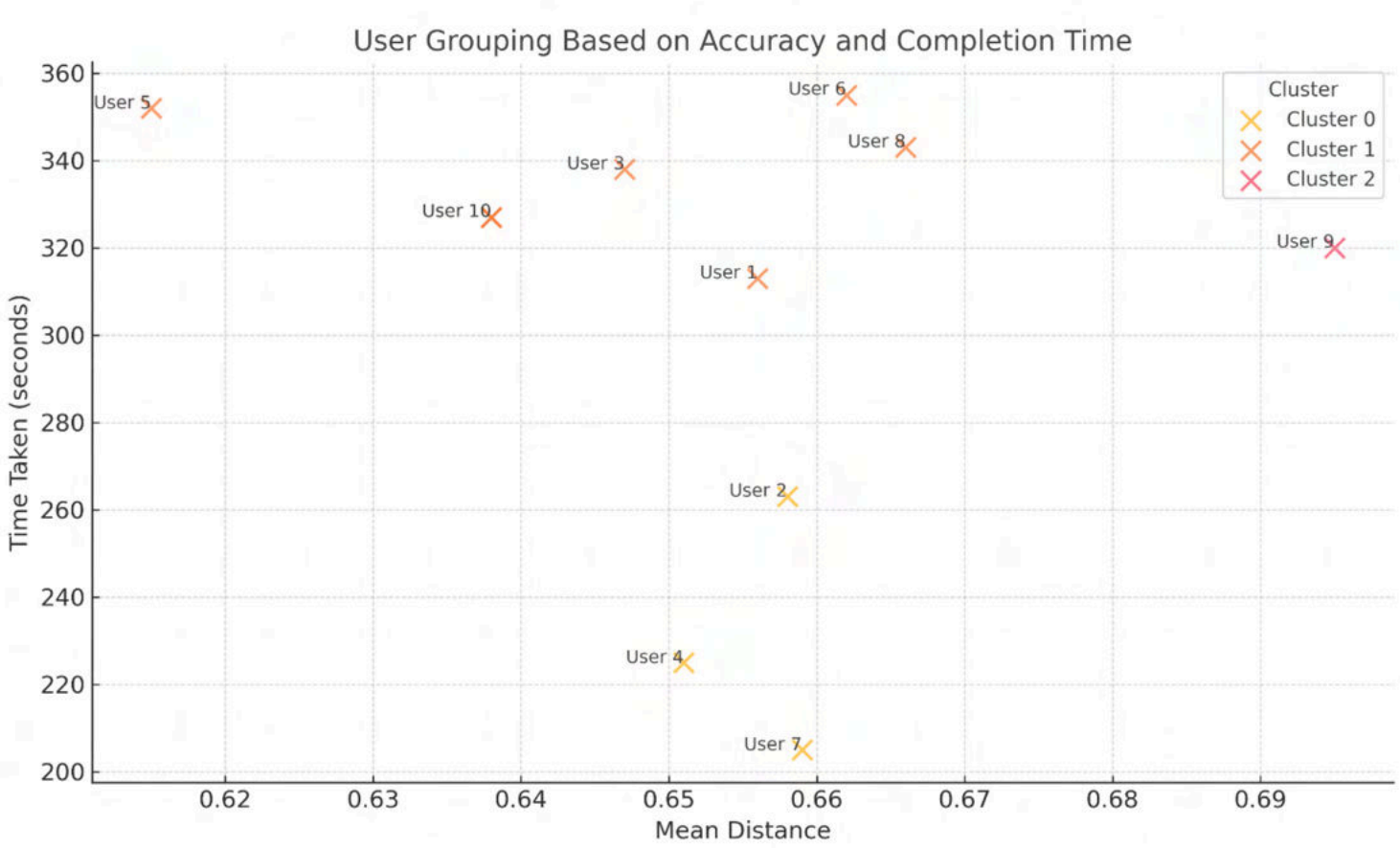
Joint Adjustment to Pre-defined Positions

Task 2

Target Alignment and Spatial Navigation

Task 3

Waypoint Navigation



The operational workflow concludes with the Waypoint Panel and Camera Views which enable users to plot precise spatial targets and visually verify their safety from multiple angles before execution. To rigorously evaluate this framework I conducted a structured user study where participants utilized these tools to navigate complex path-planning scenarios. The statistical data presented in the charts on this page summarizes the findings by correlating task completion times with collision frequency and precision metrics. This analysis reveals a significant reduction in errors and a rapid acquisition of kinematic intuition among novice users. These results validate the project's core thesis that embodied VR interaction serves as a powerful accelerator for robotics education. Ultimately JointLab demonstrates that removing the physical risks of hardware allows students to master the logic of fabrication faster and more creatively than traditional methods which effectively democratizes access to advanced manufacturing.

RECURSIVE AGGREGATIONS

iitComputational Design Lab

Year: *Spring 2020*

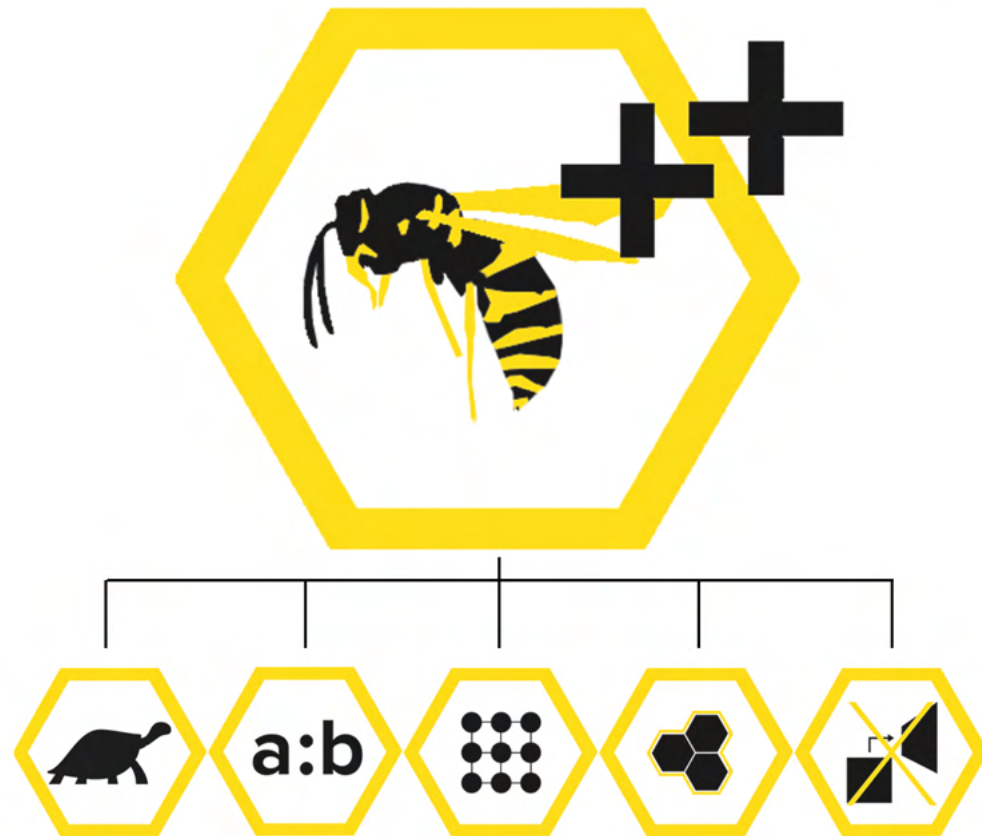
Type: *Academic Work*

Team: *Blake Hageman, Nhat Nguyen, and Nate Staniak*

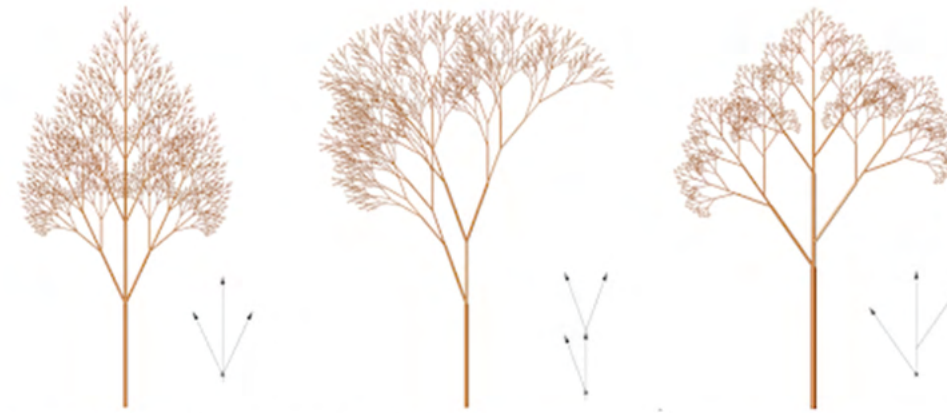
Tutor: *Alphonso Peluso*

Link to Booklet: <https://bit.ly/3Ylp9sm>

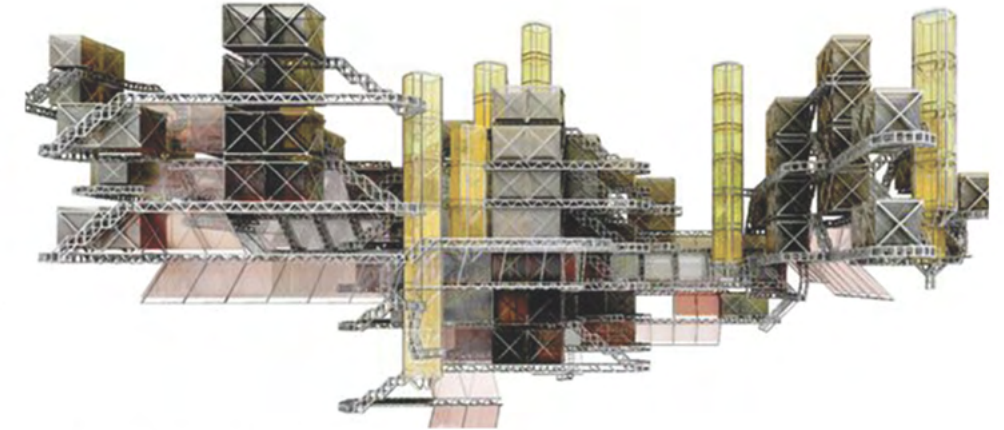
This study examines the computational control and manipulation of recursive structures through discrete element aggregation. The main product of the research is a suite of Python-coded components called Wasp++, built on top of Wasp for Rhinoceros/Grasshopper, to study alternative controls for aggregations. The core functions for Wasp++ include L-system aggregations, hierarchical part definitions, and translating aggregations to graphs. The added functionality for controlling discrete element systems suggests several practical applications for future work, particularly in manufacturing and sequencing the construction of system parts.



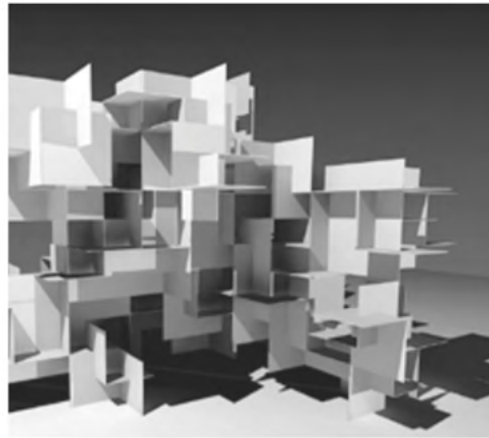
Wasp ++Plugin Components



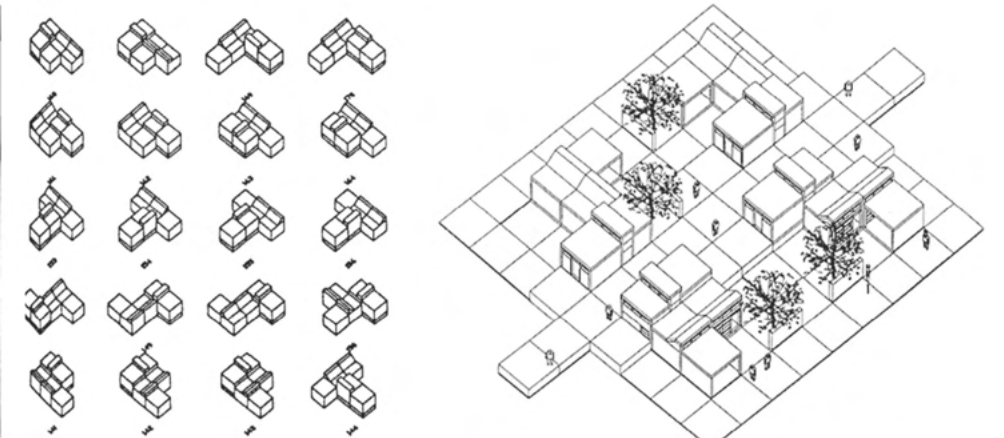
Fractal Trees, Generative Landscapes



L-Systems in Architecture



Generative Algorithmic Techniques for Architectural Design



Analogical and Inductive Reasoning in Architectural Design Computation

Generative design, recursion, and shape grammar systems are attempts to define the behavior of large systems by determining essential elements and the rules by which those elements go together. L-systems and cellular automata are two of the most prominent families of these logical structures. L-systems, in particular, were initially developed to describe the deep structure of plants through simple recursive rules.

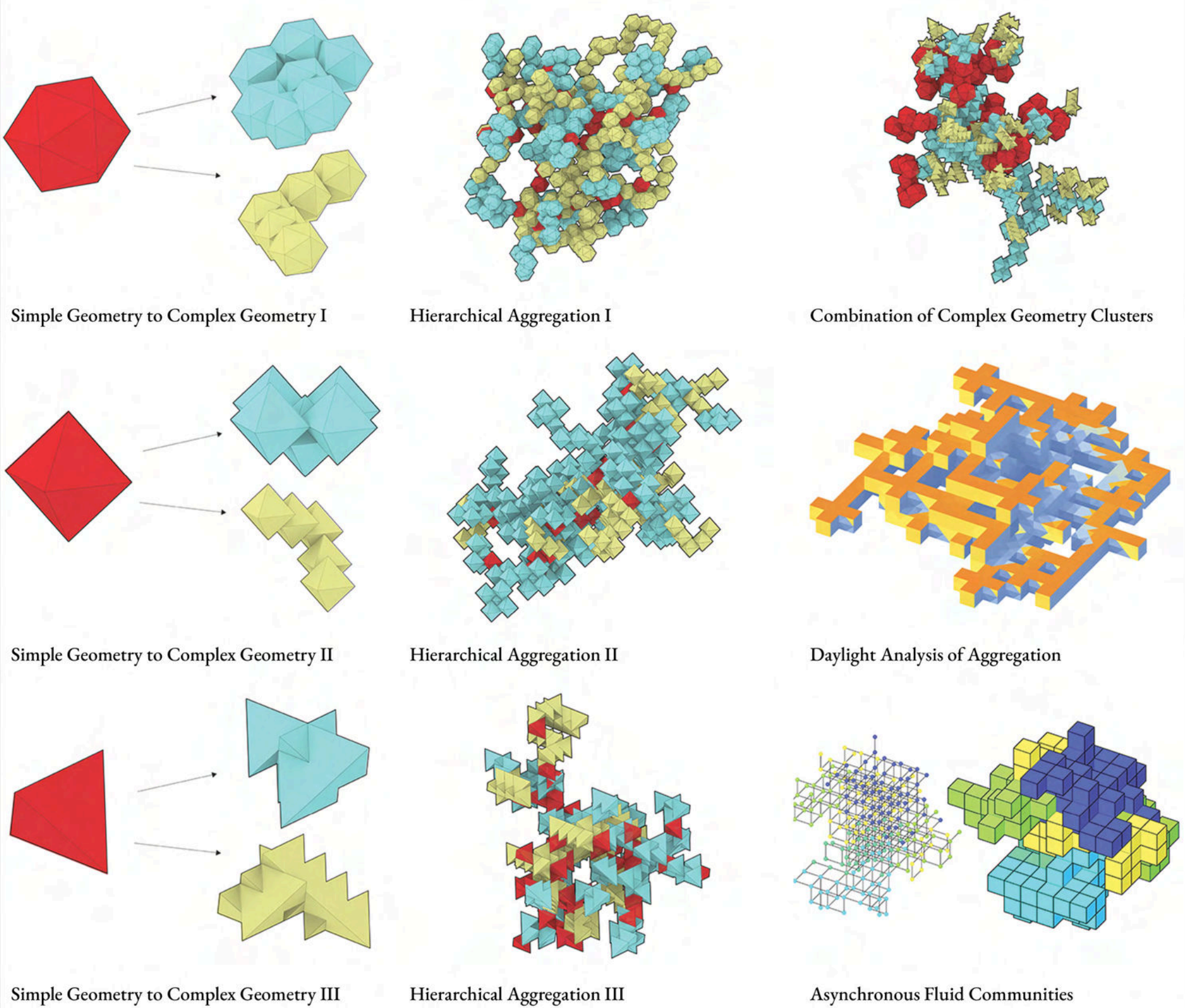
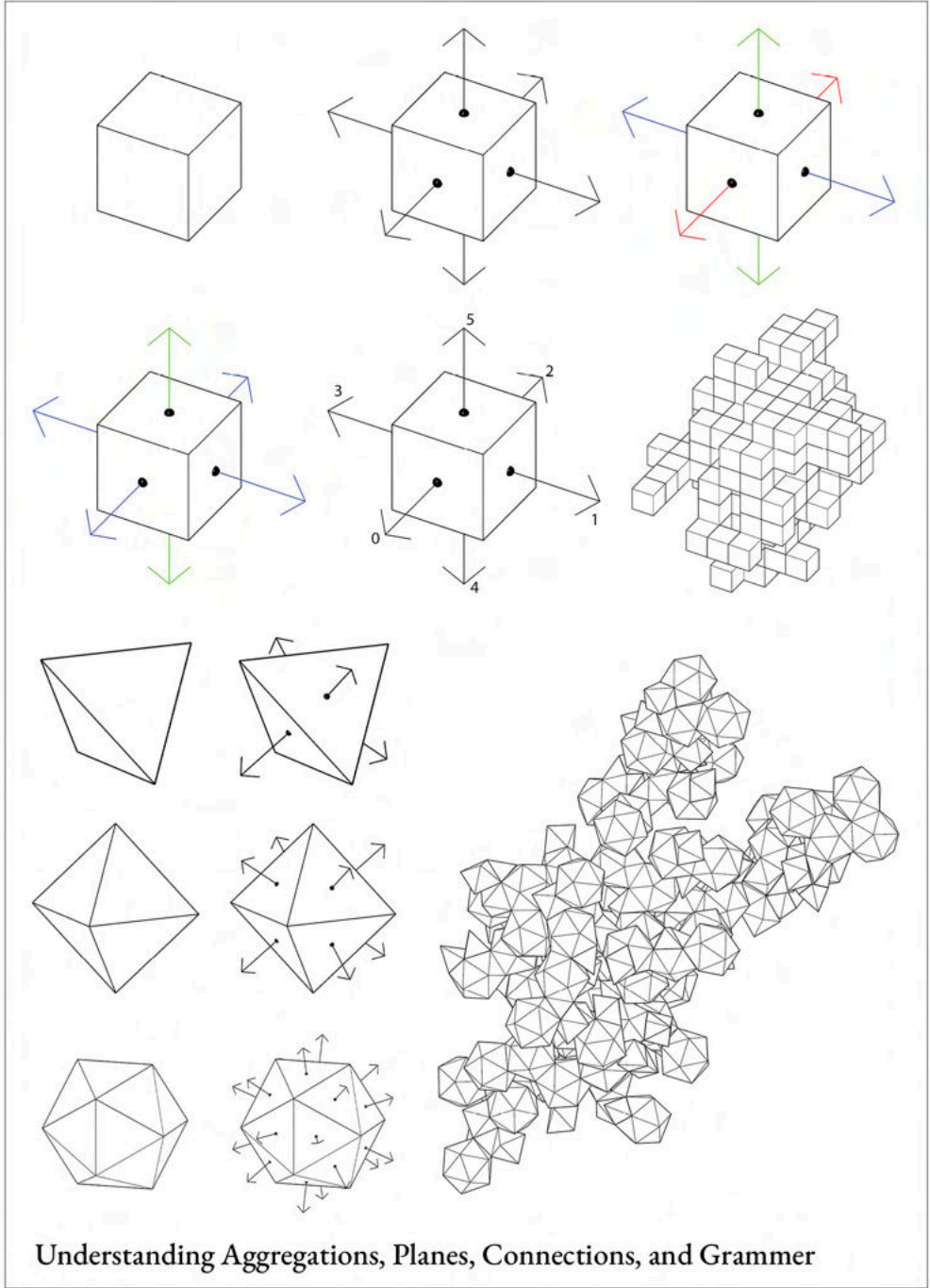
This exploration is well suited to exist means of 3D modeling through Rhinoceros and in its-suite algorithmic modeling environment Grasshopper. Defining a robust system with elements and relationships is the core task in these spaces for the research, so Wasp, an open-source plug-in for Grasshopper built by Andrea Rossi, was used as the foundational system for exploring aggregations.



System Environment: Rhino, Grasshopper, Wasp, and GhPython

The signature function of Wasp++ is a translation of the Python Turtle graphics library into a deterministic aggregation function. Like a traditional Turtle, which moves a point incrementally in 2D or 3D space to draw lines, shapes, and complex patterns, the Turtle Aggregation component treats each discrete element of the aggregation as a point, which the turtle adds incrementally according to a series of instructions. Since aggregations are already defined with a pool of rules for possible connections, the instruction grammar is a stack structure that references the rules by index, with additional symbols for branching and moving the cursor. Calling a rule implements that connection-to-connection addition to the aggregation at the cursor location.

With a deterministic control in place, the instruction set can be created with an L-system generator to create complex recursive structures with a minimal set of inputs. These aggregations can imitate well-known L-systems patterns or generate unique 3D geometries.



In addition to new modes of managing clusters, several supplemental methods were built that augment or add functionality to the existing Wasp Stochastic Aggregation component. I was in charge of writing the Pseudocode, Testing the code, and creating an example file for Ratio Aggregation, which uses the basic framework of the utterly random aggregation method, but controls for the proportions of each part in the aggregations. The part with a count farthest below the target number based on the ratio is added next. This gives more control over the aggregation randomness by controlling which part is added next and determining where it is added. I also looked primarily into daylight analysis and finite element analysis; both methods proved too slow and cumbersome to adjust an aggregation shape iteratively. A single iteration for daylight analysis with Grasshopper plug-in Ladybug, even with low-resolution information, could take several seconds. When aggregations have parts in the hundreds or thousands, such approaches become quickly untenable. Potential areas of further study have been described above but again include performance optimization, dynamic part clustering, aggregation tiling, and sequencing. Future uses of the plugin were analyzed to understand further implementations.

Robotics

Carnegie Mellon Computation

Year: *Fall 2023*

Type: *Academic Work*

Team: *Myles Sampson, Sanchit Agrawal & Sejal Jain*

Tutor: *Joshua Bard & Jingyang Liu*

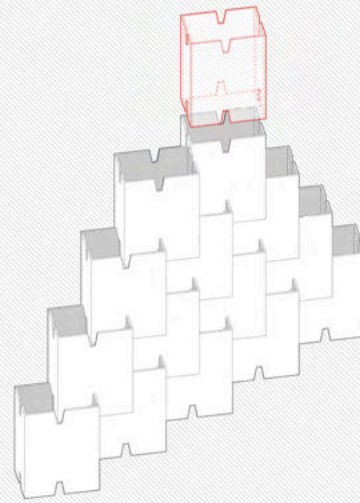
The course offered a hands-on introduction to integrating industrial robots into architecture and construction practices, providing practical insights into various aspects such as online programming, offline programming, simulation, end-of-arm tool development, workcell development, and interfacing with peripheral sensors.

In our project, we chose to concentrate on color-detected assembly tasks. Initially, we designed three distinct module types with differing cuts for assembly and various colors, while maintaining consistent overall dimensions. Leveraging OpenCV integrated into Grasshopper through GhHowl, we successfully detected the colors of each module. Subsequently, each color was assigned a specific organizational pattern, dictating a unique stacking method. The stacking logic was then implemented within RobotStudio, establishing a seamless connection between OpenCV, Grasshopper, and RobotStudio.



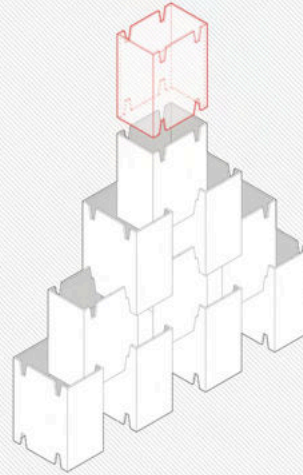
Red Color Assembly

Color: Blue



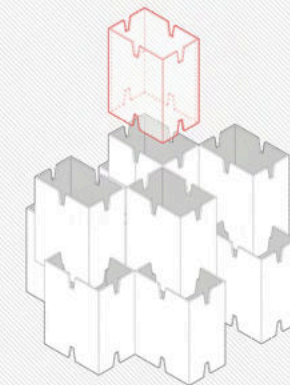
Assembly 01

Color: Red



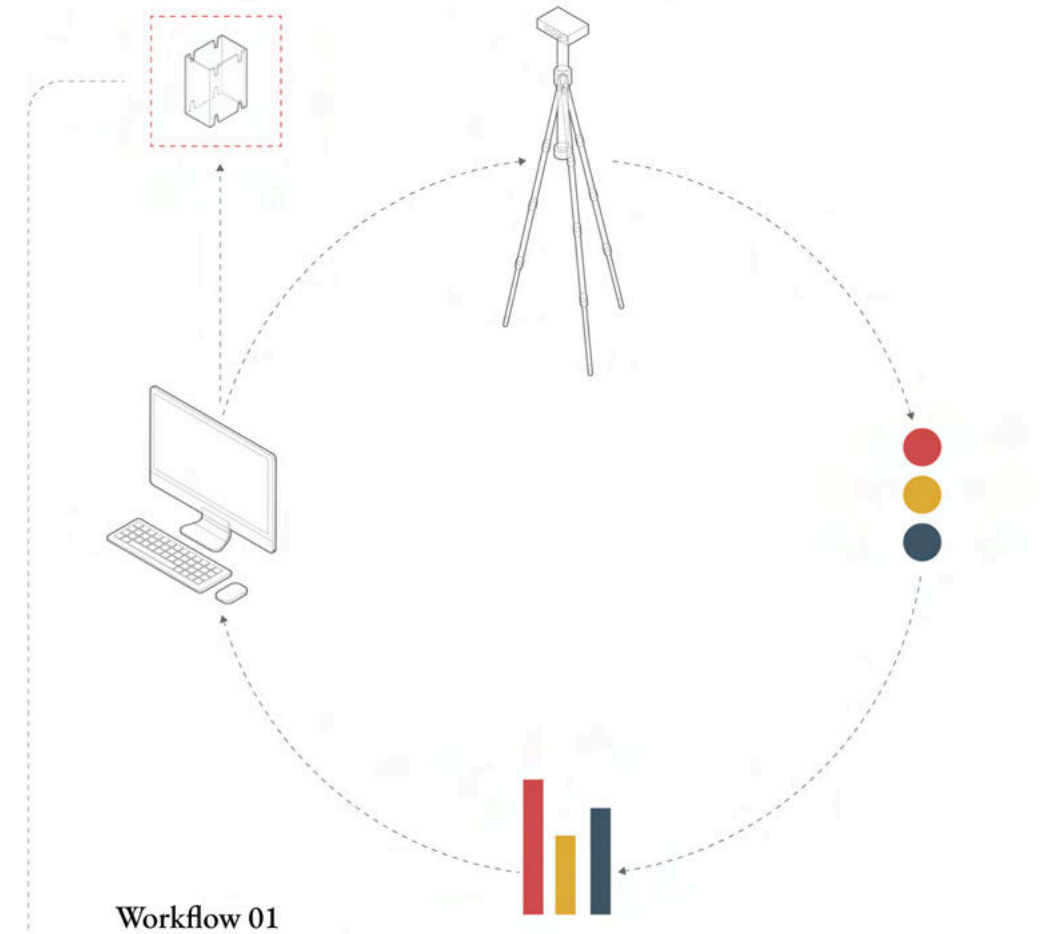
Assembly 02

Color: Yellow



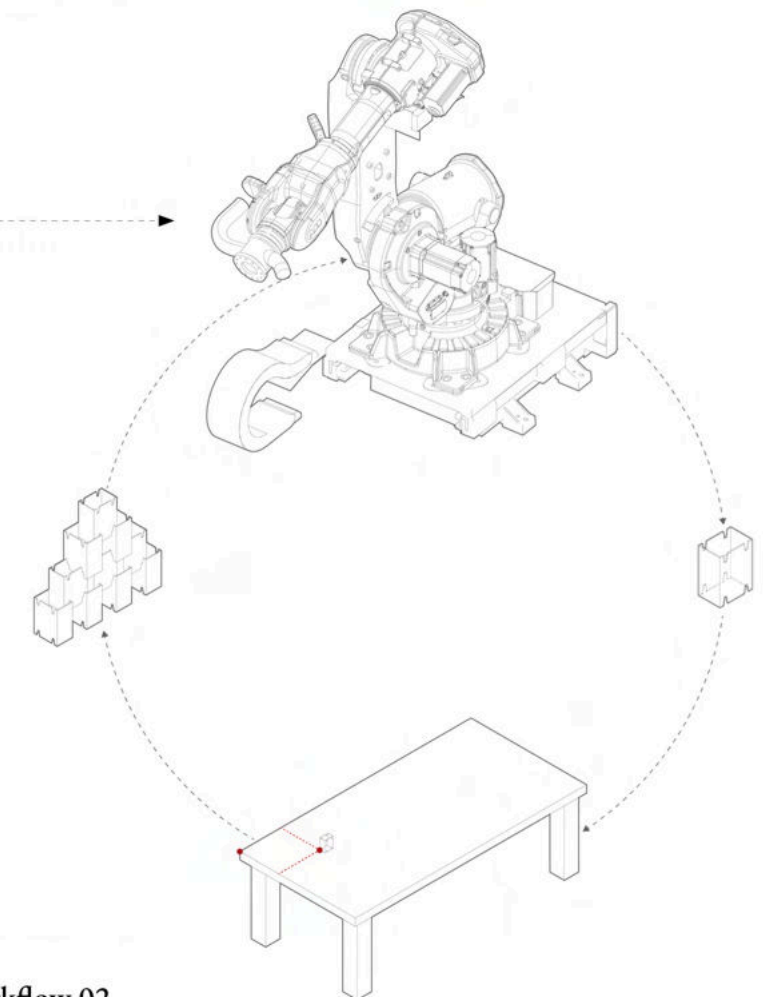
Assembly 03

Color Detection | Using OpenCV & Grasshopper

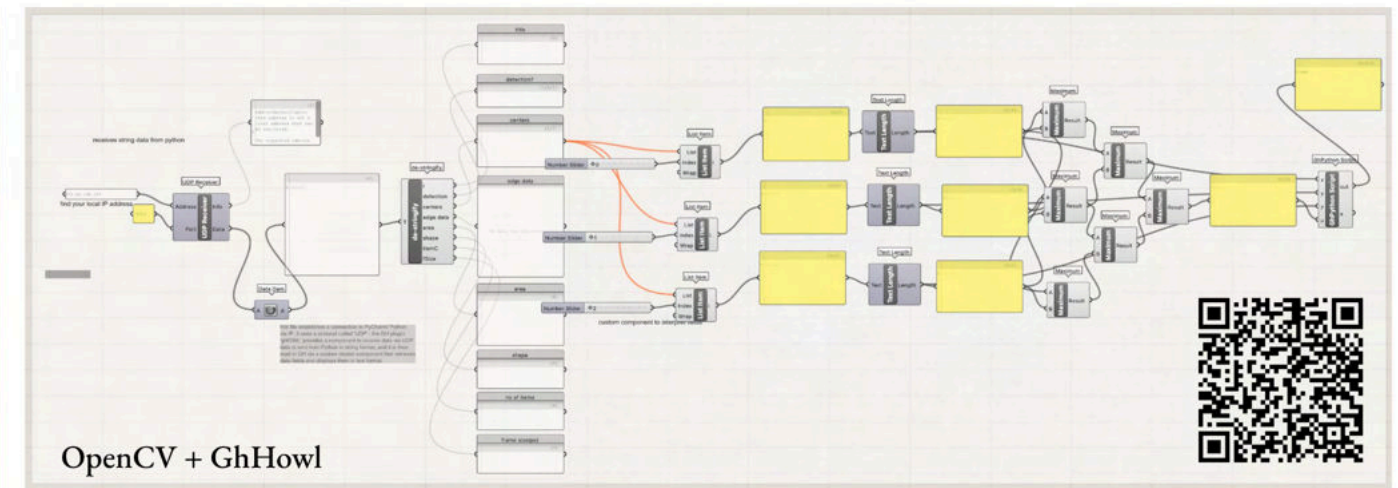
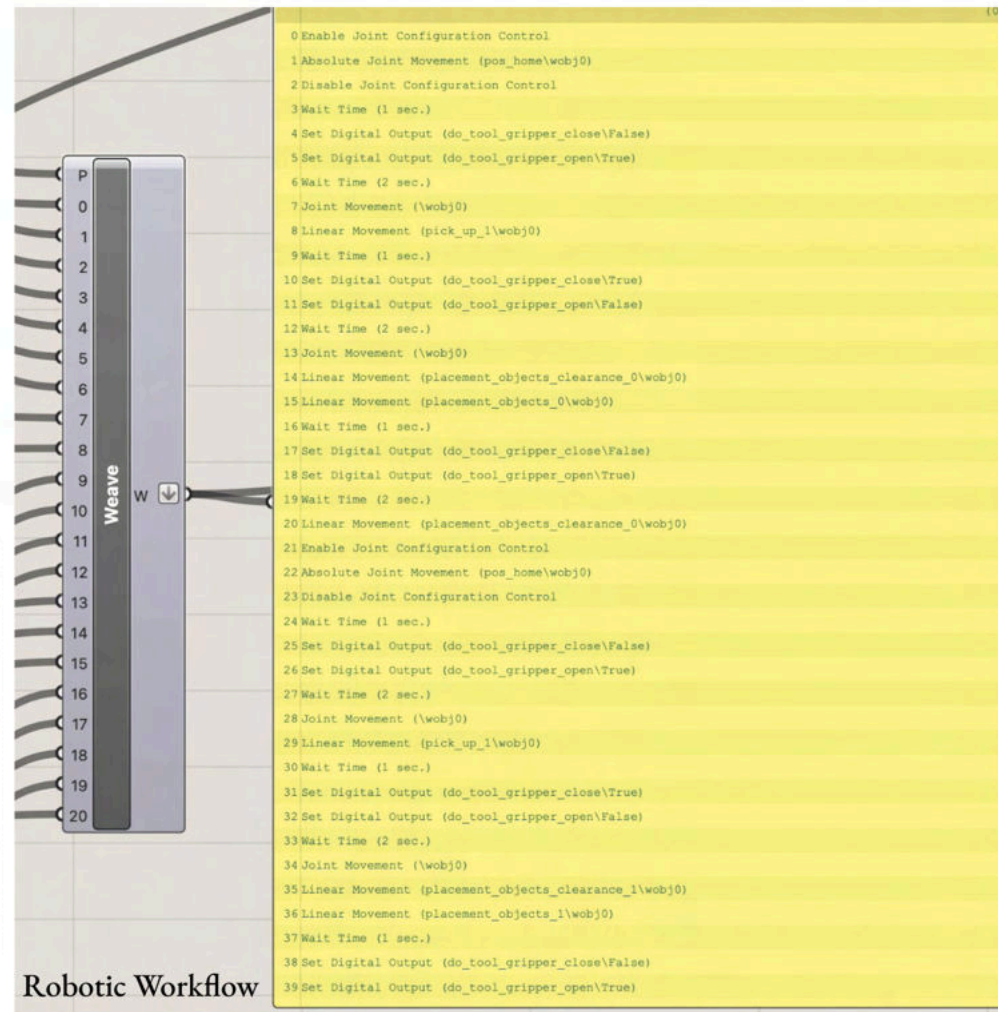
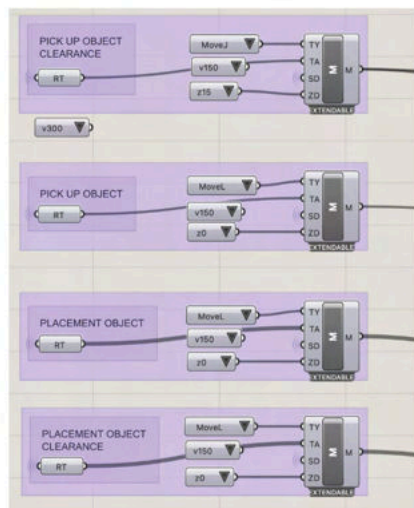


Workflow 01

Pick & Place | Grasshopper & RobotStudio

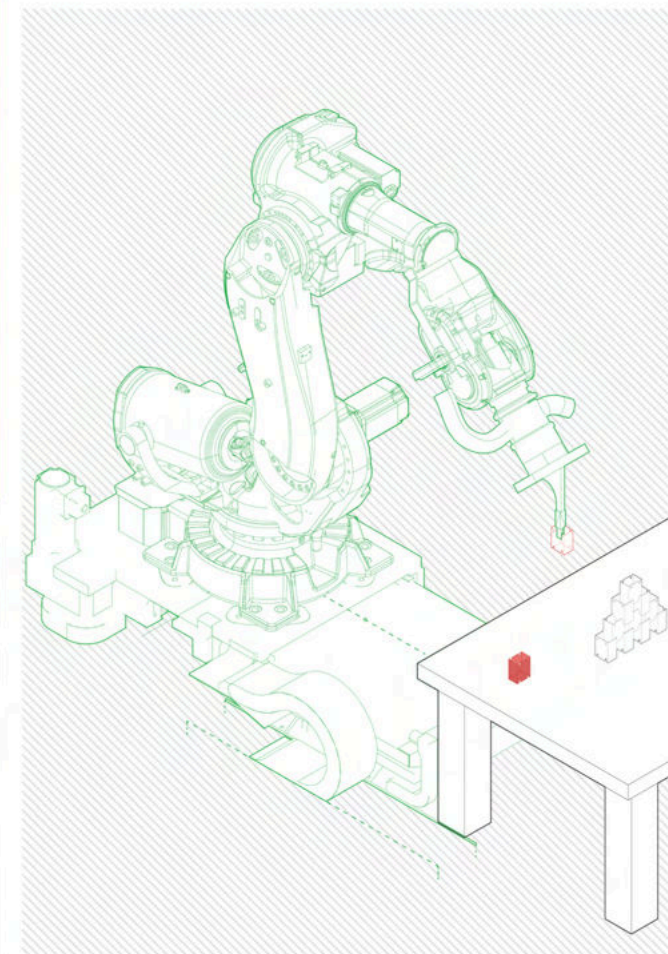


Workflow 02

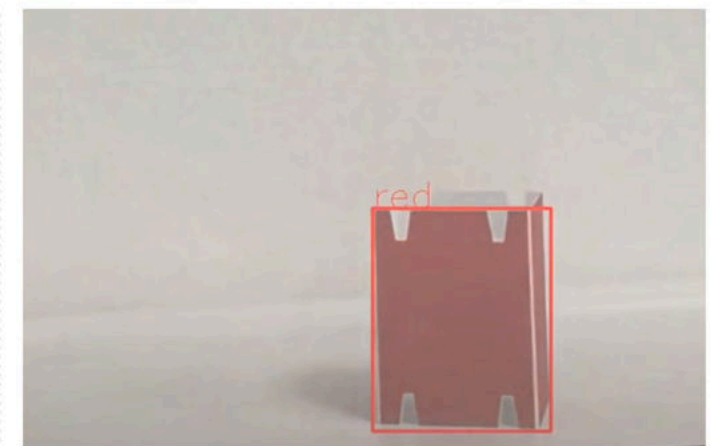


End Tool - Pneumatic Point Gripper

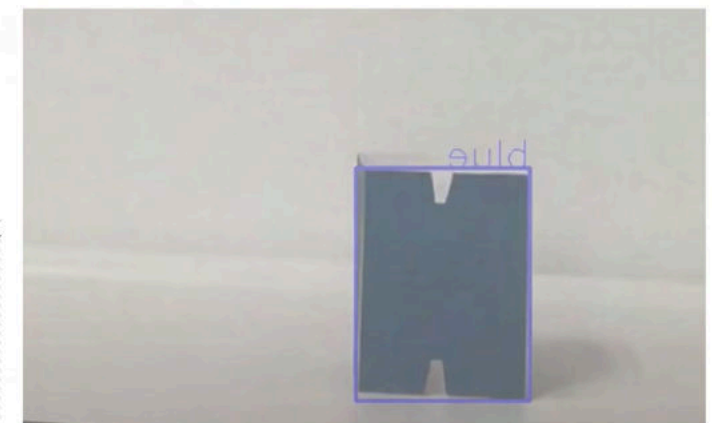
Robotic Workflow



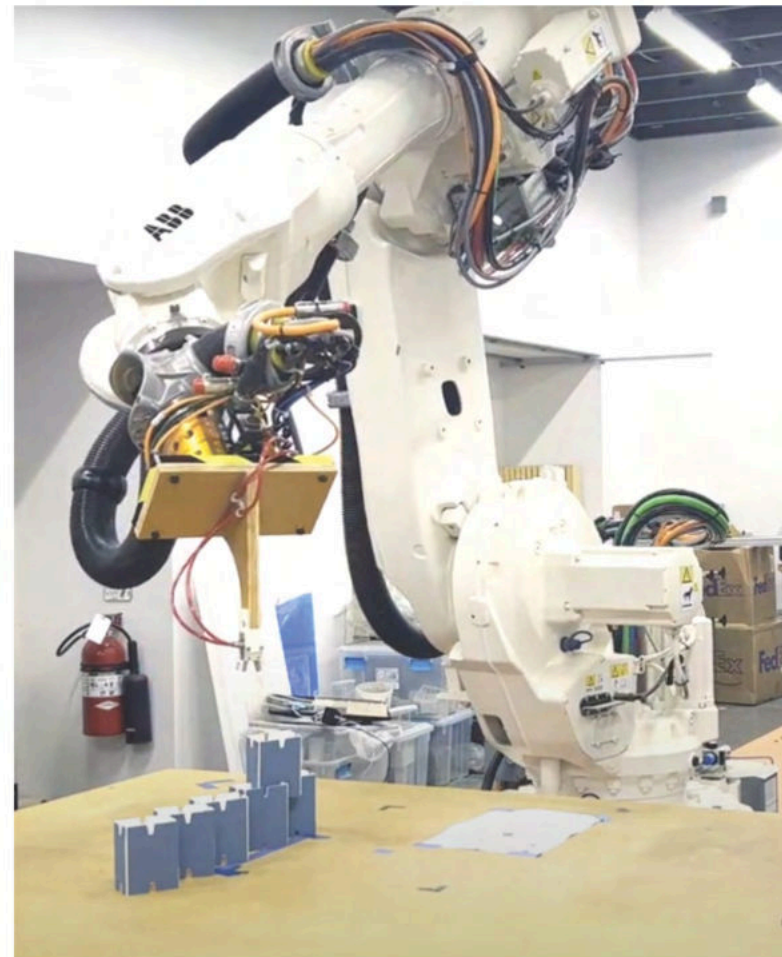
Pick and Place



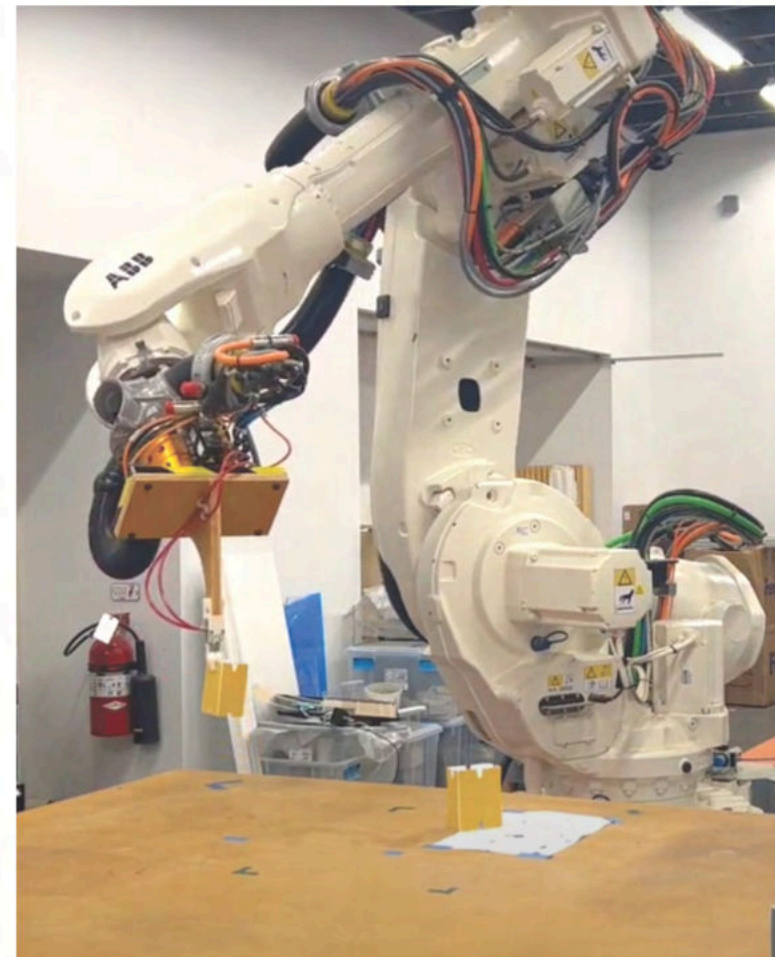
Detection 01



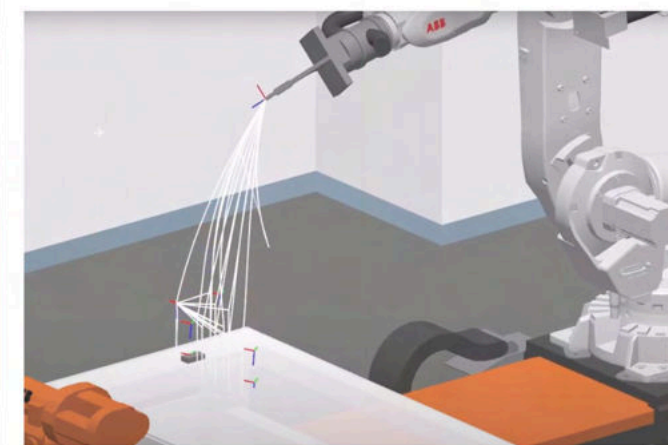
Detection 02



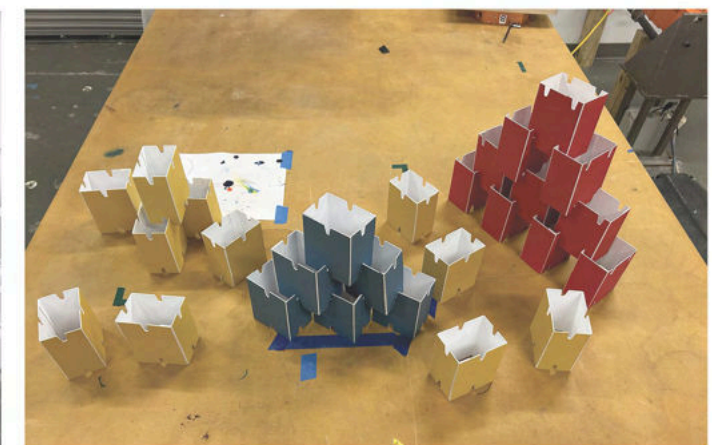
Blue Color Operation



Yellow Color Operation



Simulation



Work Station

Silver Clouds

Carnegie Mellon Computation

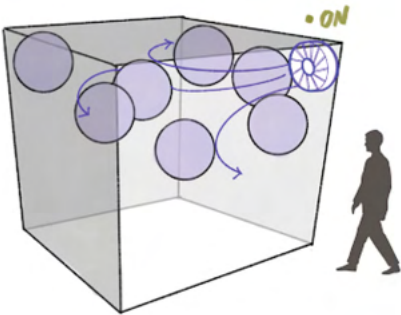
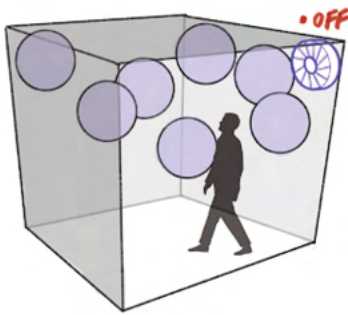
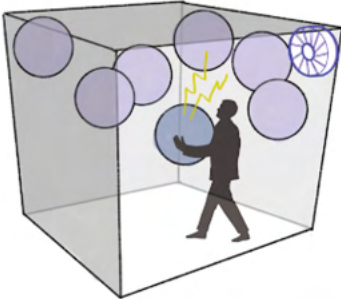
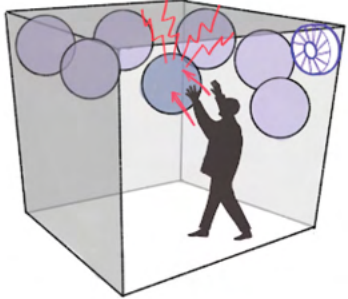
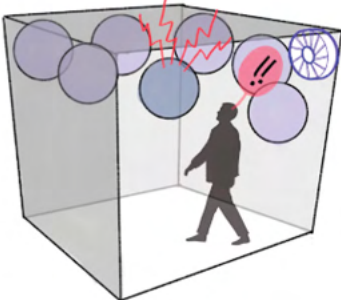
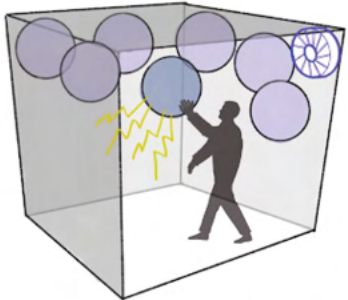
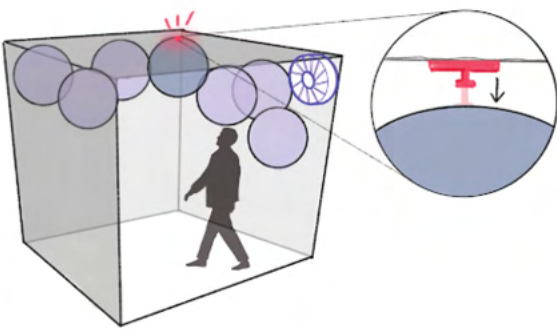
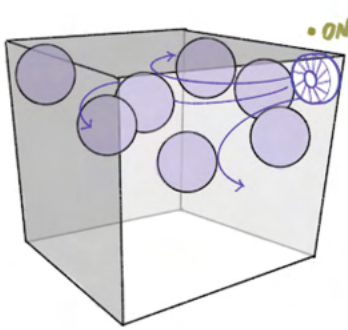
Year: *Spring 2024*
Type: *Academic Work*
Team: *Sonia Prashant, Shannon Lee & Khevna Modi*
Tutor: *Ensar Temizel & Paul Pangaro*

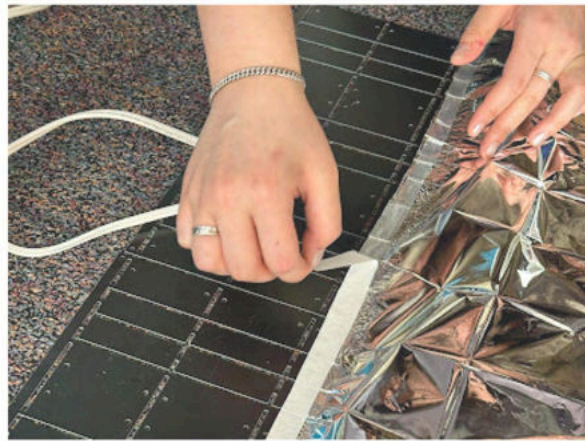
As the teaching assistant for this course, I collaborated closely with Prof. Ensar and Paul, taking charge of the Arduino programming essential for this interactive installation.

The installation draws inspiration from **Silver Clouds** at the Andy Warhol Museum in Pittsburgh, inviting students from the College of Fine Arts to engage with the piece. Within an enclosed space, fans activate when unoccupied, causing the balloons to move and enticing visitors to enter. Once someone steps inside, the fans turn off, allowing the visitor to interact directly with the balloons, which emit sounds based on the force of interaction—the greater the impact, the louder the sound produced. This feature encourages users to adjust their touch to produce softer pitches. Additionally, certain enclosure points trigger solenoids that push the balloons back and reset their tones upon contact. Each balloon is equipped with an Arduino, powered by a battery and preloaded with code for remote operation, ensuring a dynamic and responsive experience.

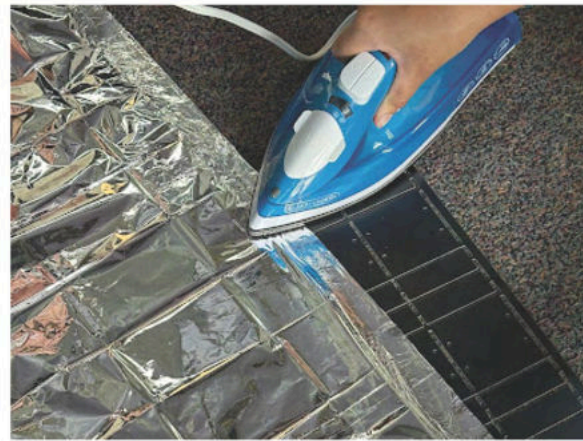


Final Installation

1. 	2. 
On his way to class, Ethan hears the fan whirring, and buzzer sounds emitting from these gently wafting balloons.	When Ethan steps into the enclosure, the fans stop.
3. 	4. 
Ethan plays with the balloon by gently pushing it. It emits a regular pitch tone.	Out of curiosity, he hits it harder, and the balloon emits a higher, more piercing pitch.
5. 	6. 
Sensing something is wrong, Ethan quickly adjusts his behavior and starts playing with the balloon more gently.	After some time, it calms down and emits a lower pitch sound.
7. 	8. 
Ethan realises that when he hits the balloon towards the enclosure's ceiling at certain points, the enclosure hits the balloon back.	After some time, Ethan leaves to head to class. When he leaves, the fans whirr on again.



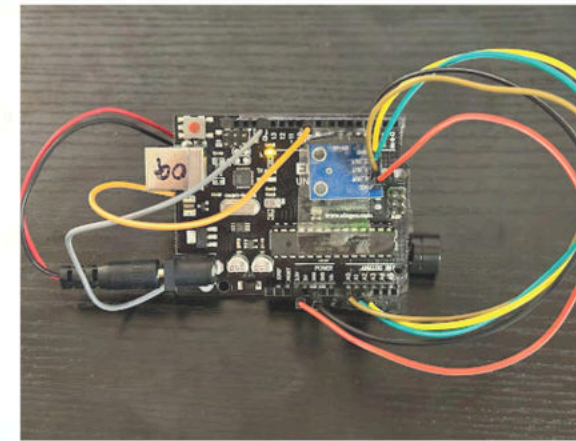
Adhesive towards the edges



Balloon sealed with iron



Helium added in bursts

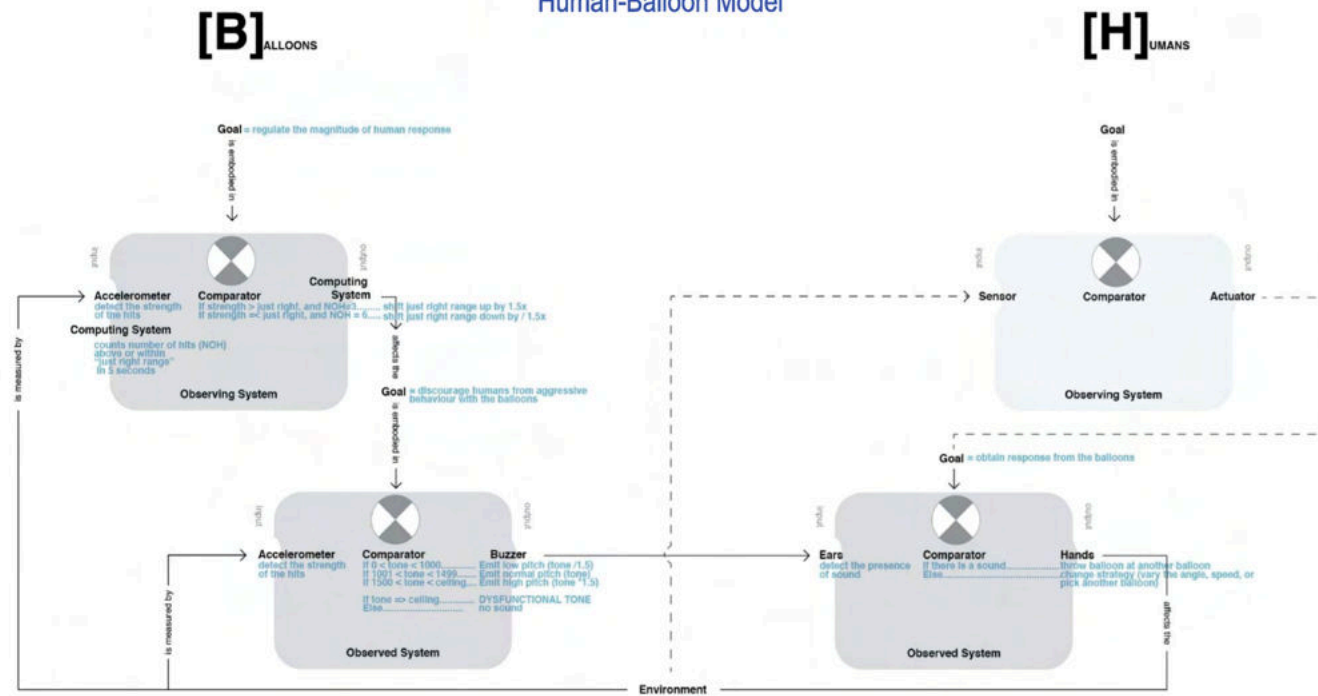


Assembled Arduino System

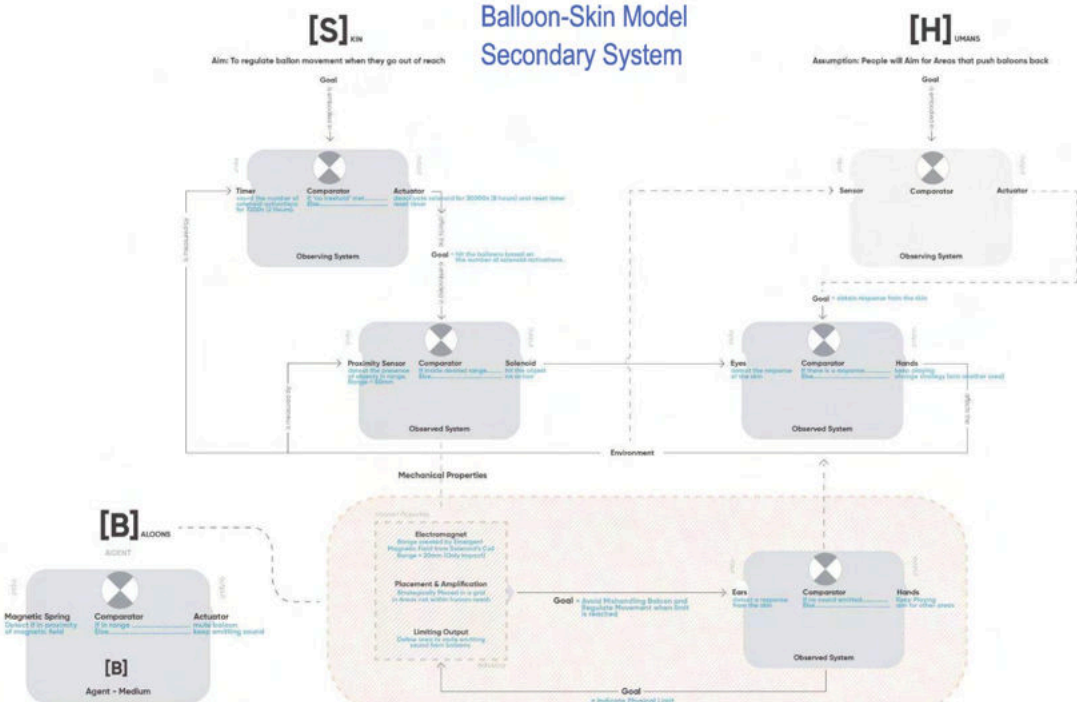


Balloon with Arduino taped inside

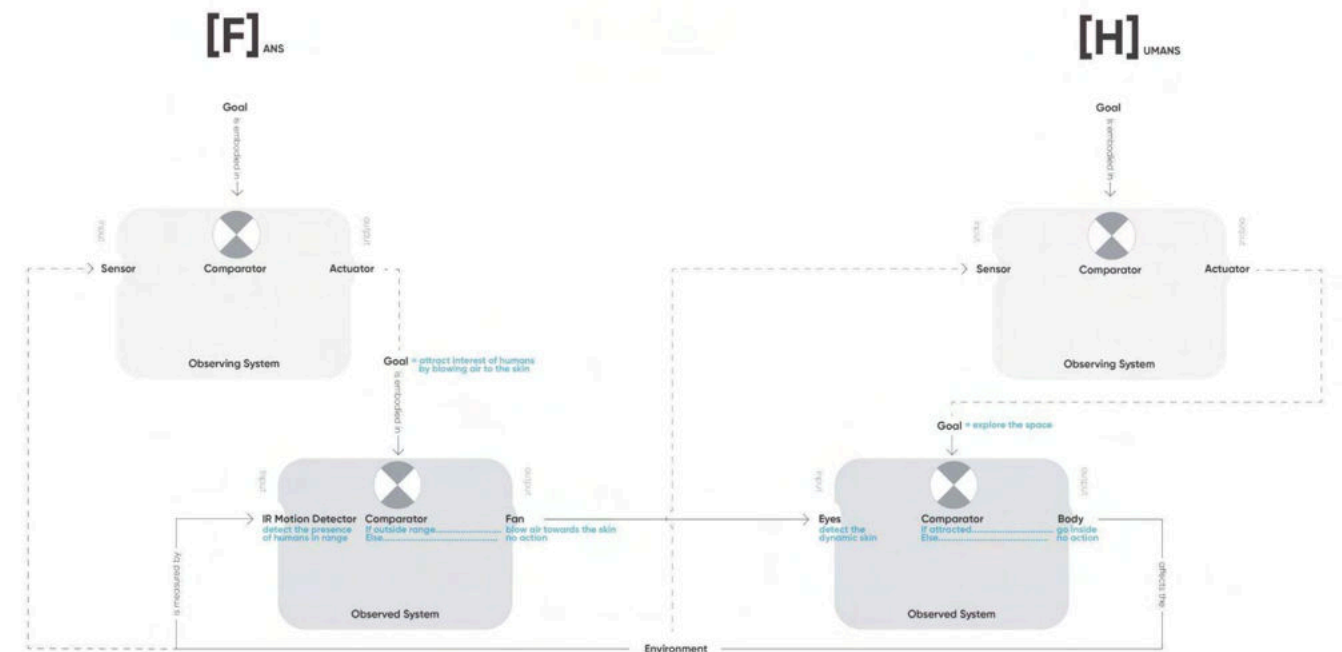
Human-Balloon Model



Balloon-Skin Model Secondary System



Fan-Skin Model



Arduino Code

```
void setup() {
  Serial.begin(9600);
  pinMode(xPin, INPUT);
  pinMode(yPin, INPUT);
  pinMode(zPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
}

void loop() {
  // Read raw data from the accelerometer
  int xRaw = analogRead(xPin);
  int yRaw = analogRead(yPin);
  int zRaw = analogRead(zPin);

  // Calculate g-force
  float xG = (xRaw - zeroG) / scale;
  float yG = (yRaw - zeroG) / scale;
  float zG = (zRaw - zeroG) / scale;
  float gForce = sqrt(xG * xG + yG * yG + zG * zG);

  // Map g-force to buzzer tone
  float toneValue;
  if (gForce < 3.6) {
    toneValue = map(gForce, 2.4, 3.6, 10, 40);
  } else if (3.61 <= gForce && gForce <= 4.3) {
    toneValue = map(gForce, 3.61, 4.3, minTone, maxTone);
  } else {
    toneValue = maxTone; // Limit the tone value at the max tone
    for g-force above the max(in the current case 4.3)
  }

  // Timing to create tone every 500 ms
  if (millis() - lastToneTime >= toneInterval) {
    lastToneTime = millis();
    tone(buzzerPin, toneValue);
    delay(500); // Duration of the tone
    noTone(buzzerPin);

    // Check if tone is high and increase the high tone count
    if (toneValue > highToneThreshold) {
      //toneValue = toneValue * 5; //create a high pitched tone
      only once after impacts bigger than the threshold, cancelled out!
      highToneCount++;
    }

    else if (toneValue < highToneThreshold) {
      lowToneCount++;
    }
  }
}
```

DRAWING WITH SAND

Lo-TEK Architecture in the Desert

Year: *Fall 2022*

Program: *Resting Stop*

Type: *Academic Work*

Location: *Vilarejo Betania, Lencois Maranhenses*

Team: *Olena Prysiashniuk*

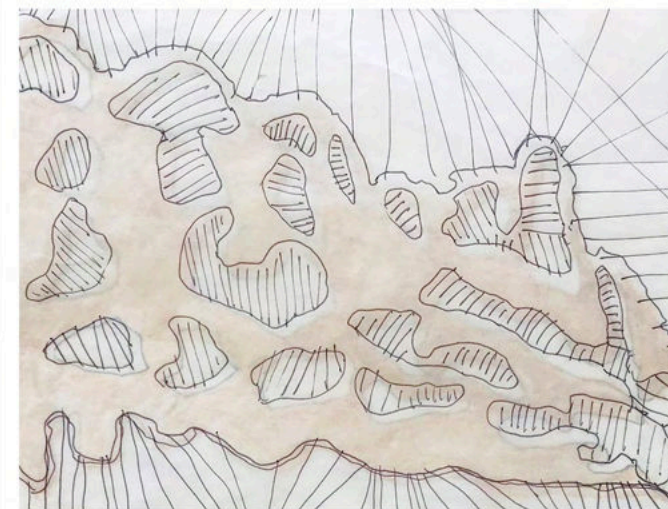
Tutor: *Gustavo Utrabo & Brianda Mireles*

Drawing on Sand proposes creating a Resting Stop within the desert of Lencois Maranhenses, Brazil, near Vilarejo Betania. The program served as a meeting point between the community and tourists for exchanging ideas. When combined with different bonding agents, we researched the basic composition of Sand to use as our aggregate for the Retaining Wall. Using bamboo and joinery techniques, we created a space within the desert that blended with this unique landscape by creating a structure suspended from this retaining wall. Building construction and experimentation were an integral part of our design process, mainly since this low-tech architecture is to be built by the community quickly without introducing foreign resources. By studying water flow in the area, we could calculate a span that would create a better connection with the change in water levels during the wet and dry seasons.



Connections to Vilarejo Betania

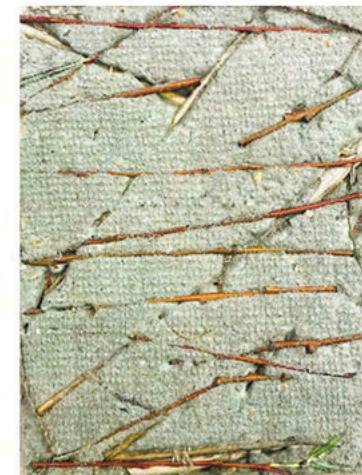
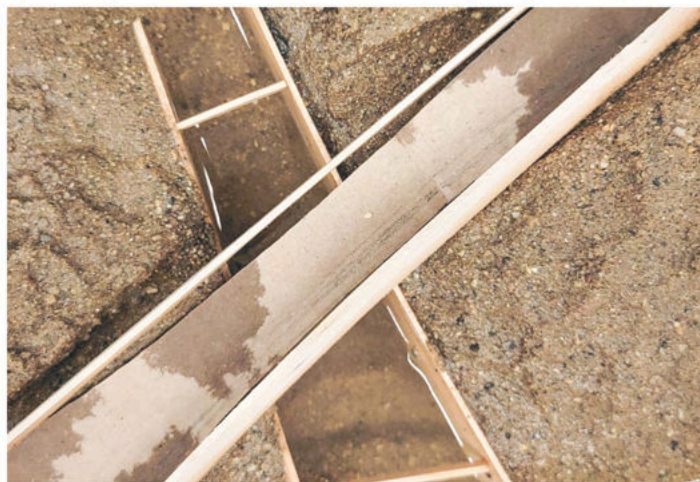
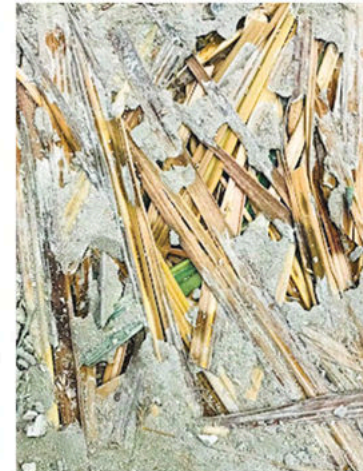
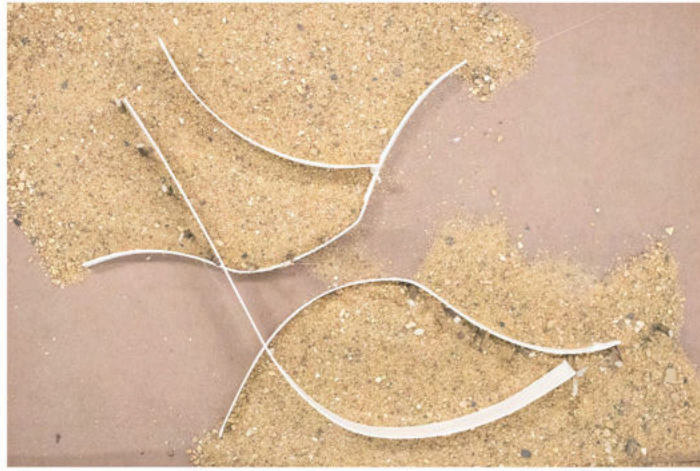
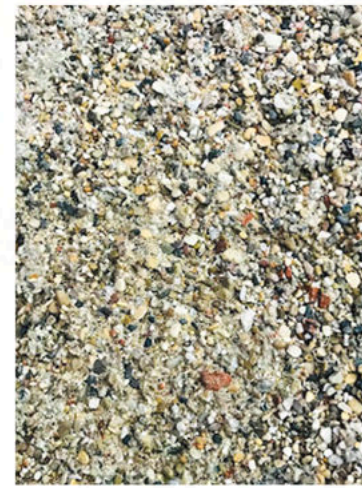
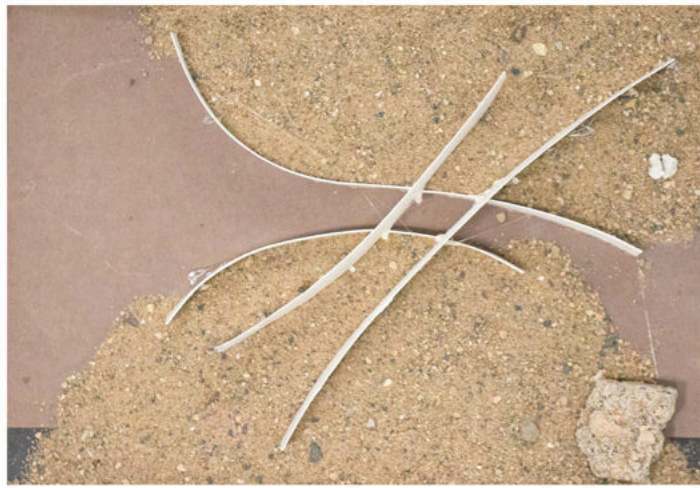




Wall Construction Exploration



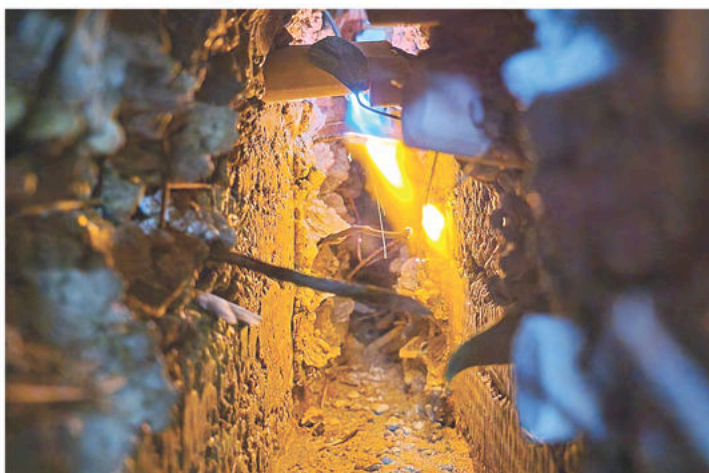
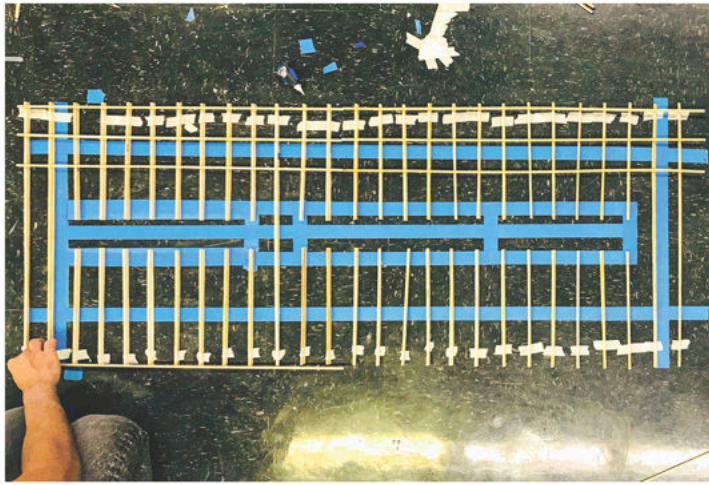
Initial Proposal



Exploration with Water

Material Palette Selection

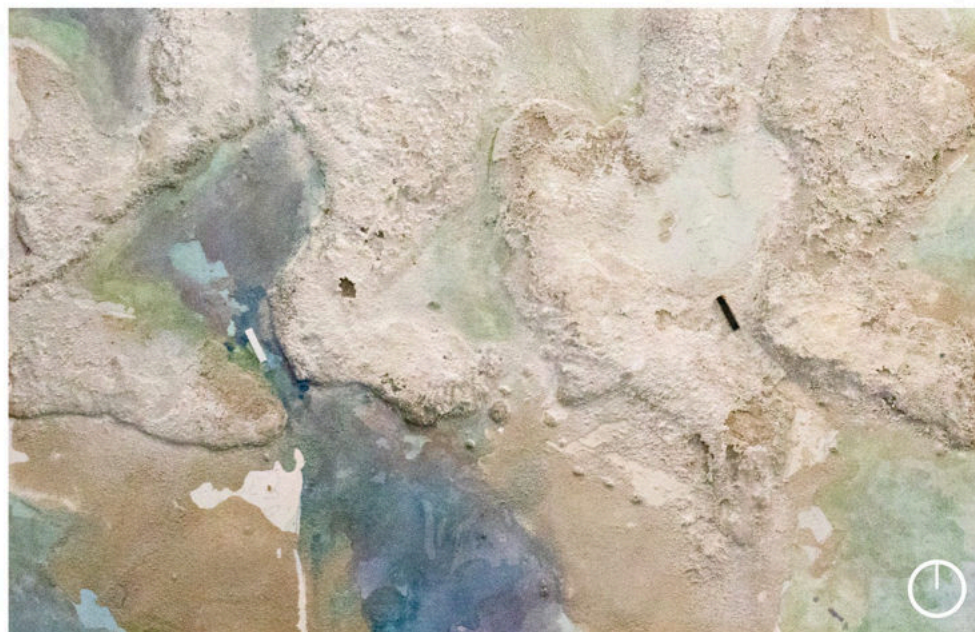
Method of Construction



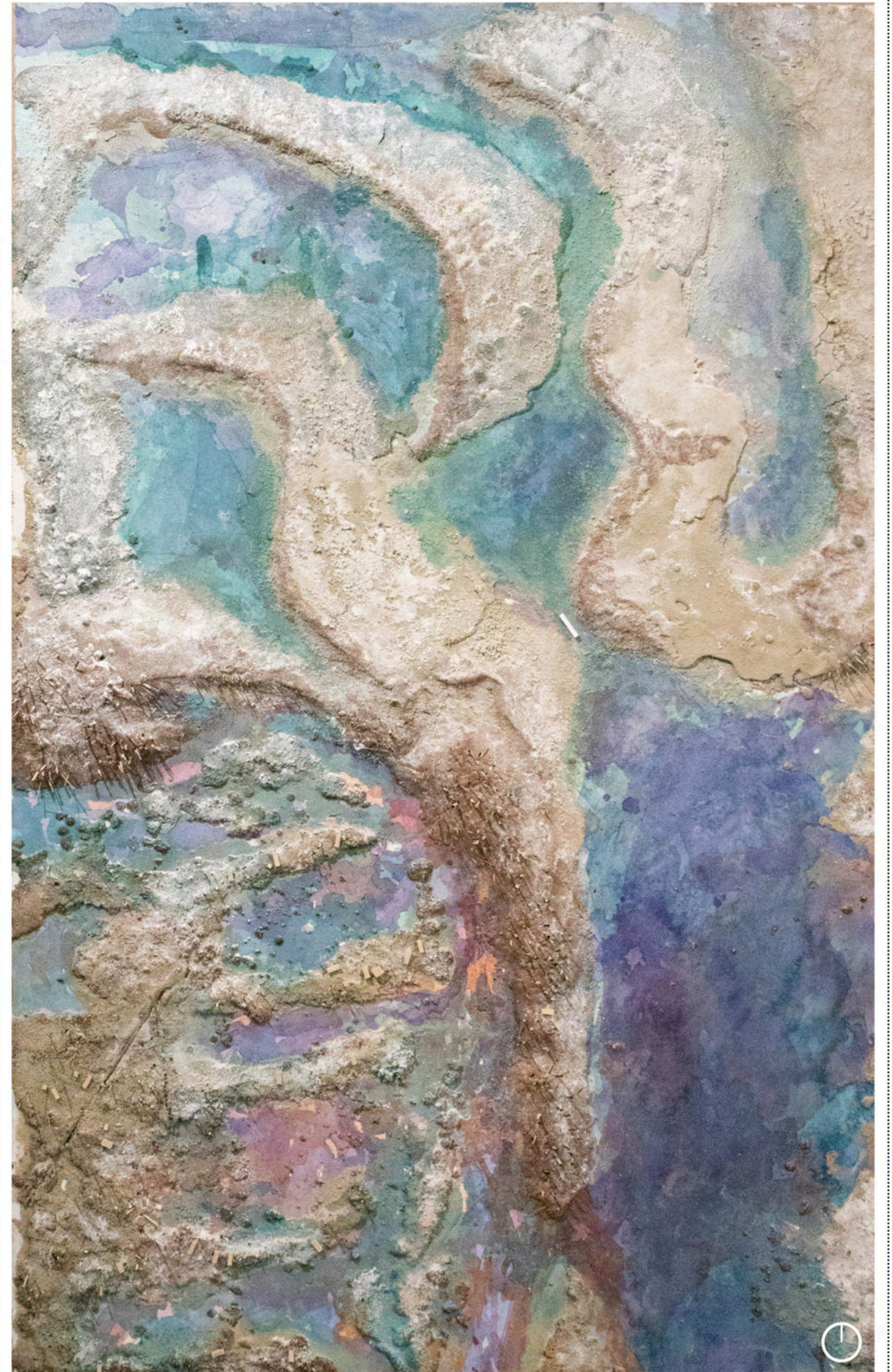
Method of Construction



Floor Plan



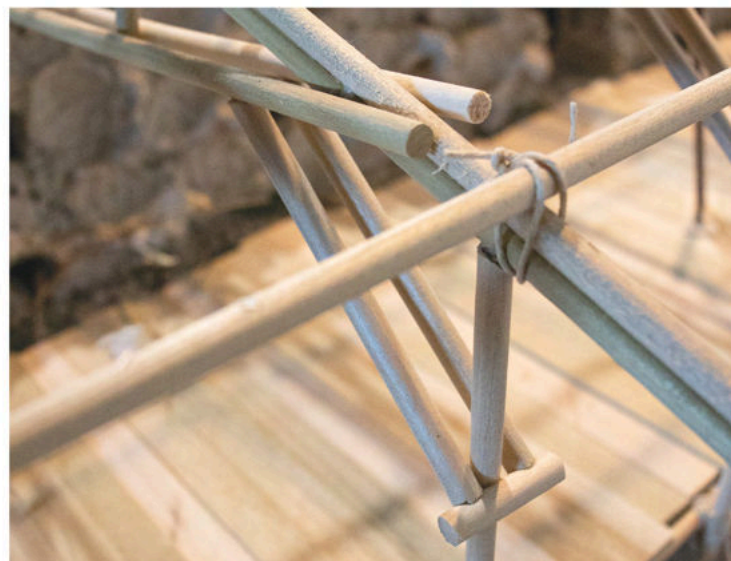
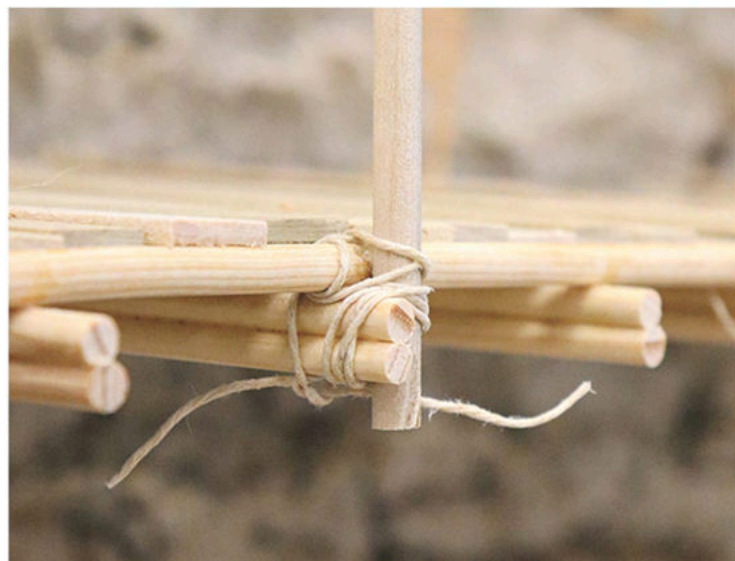
Seasonal Changes at Site



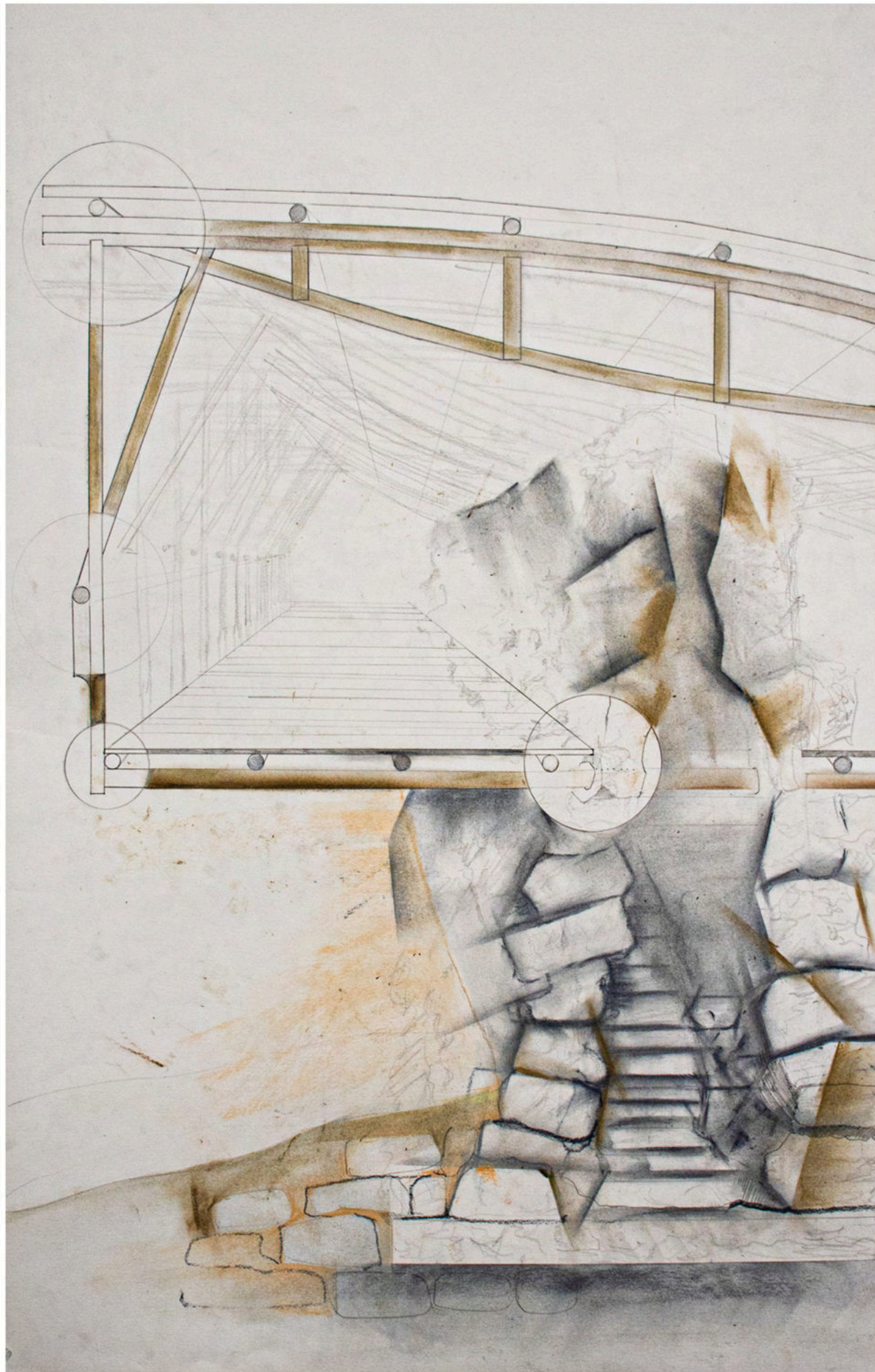
Overall Site Plan



Section



Bamboo Joinery Techniques



Wall Detail



Model Pictures



ARRIVAL

The Chicago Studio

Year: *Spring 2023*

Program: *Transportation Hub*

Type: *Academic Work*

Location: *Clark/Lake Station, Chicago, IL*

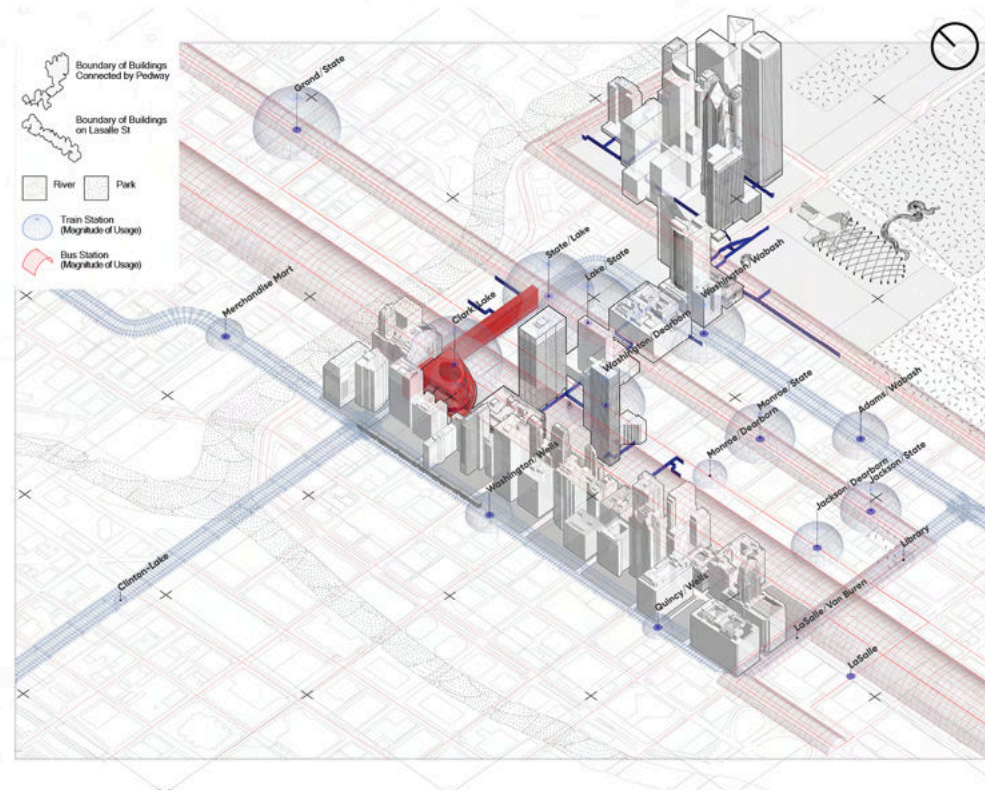
Team: *Self*

Tutor: *Joel Putnam & Philip Enquist*

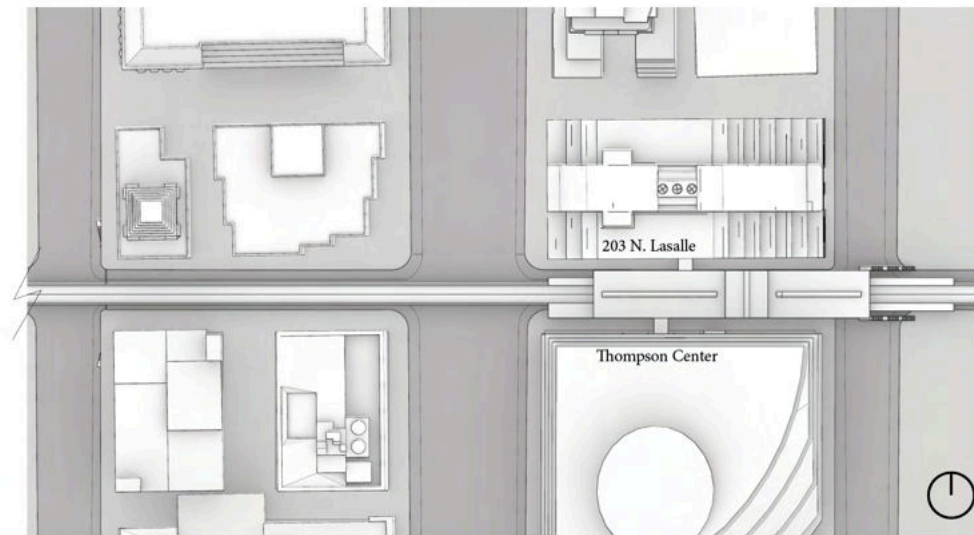
Link to Booklet: <https://bit.ly/3YffMdn>

The Clark/Lake CTA station in Chicago is a vital transportation hub in the heart of the city's Loop district. As one of the busiest stations in the city, it serves as a critical link for commuters and tourists alike. However, with its current design and layout, the station can feel congested and confined, creating a less than optimal experience for those passing through it.

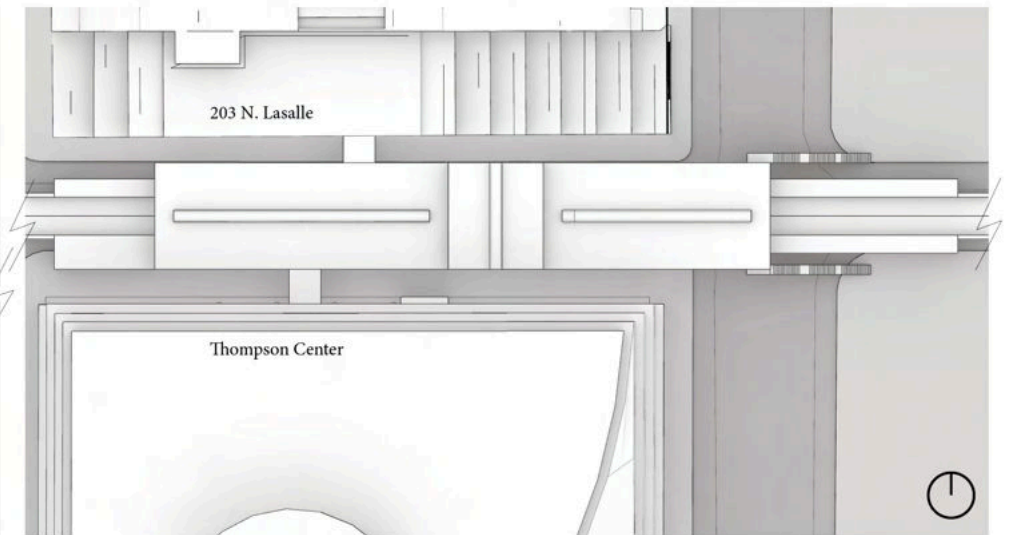
To address this issue, a new architecture project is being proposed to reimagine the Clark/Lake station, with the goal of creating a more open, lively, and functional transportation hub. The project involves setting a boundary for the station within the street itself, preventing it from spilling over into the neighboring Thompson Center and 203 North LaSalle station. Additionally, a new highline, called the Chi-Line, will be created on top of the existing L station, running the Loop.



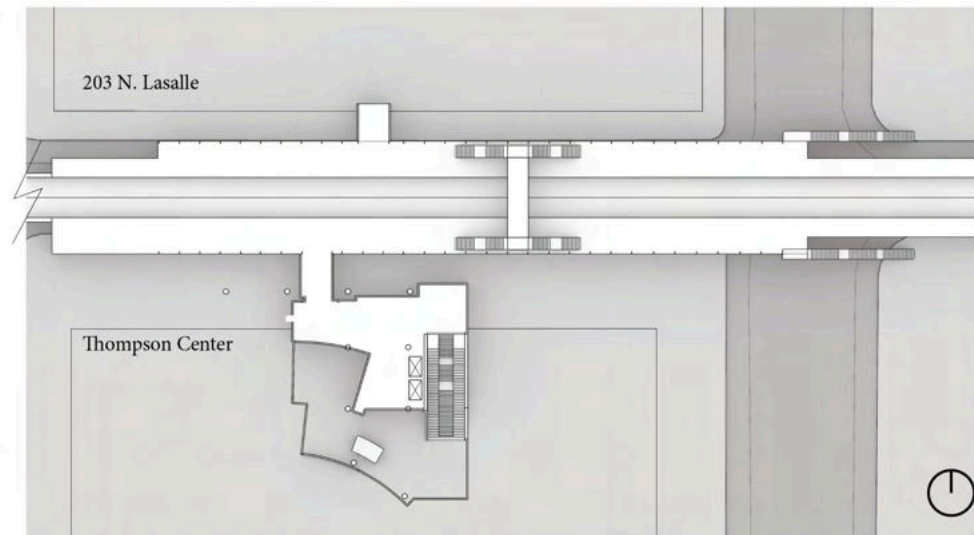
Logistics



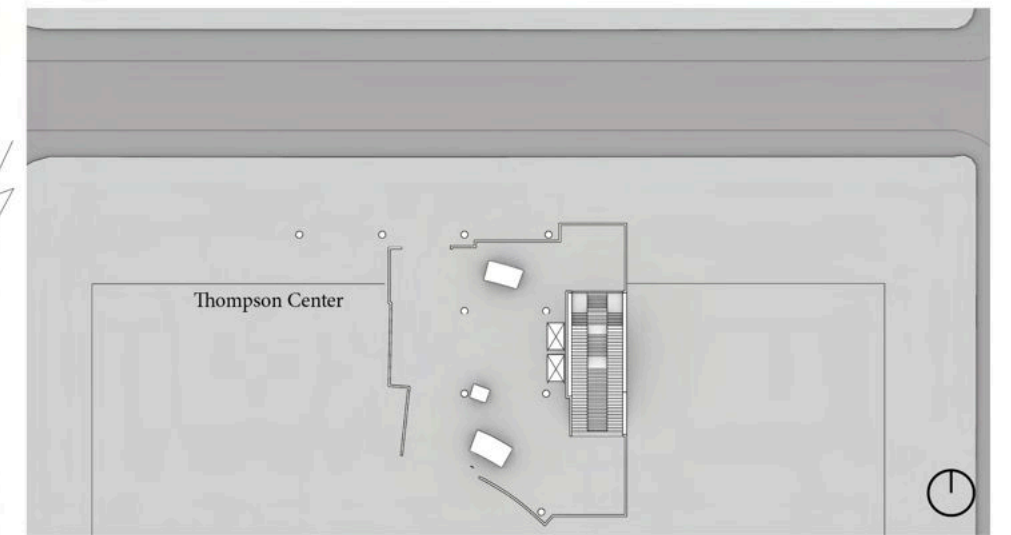
Existing Site Plan



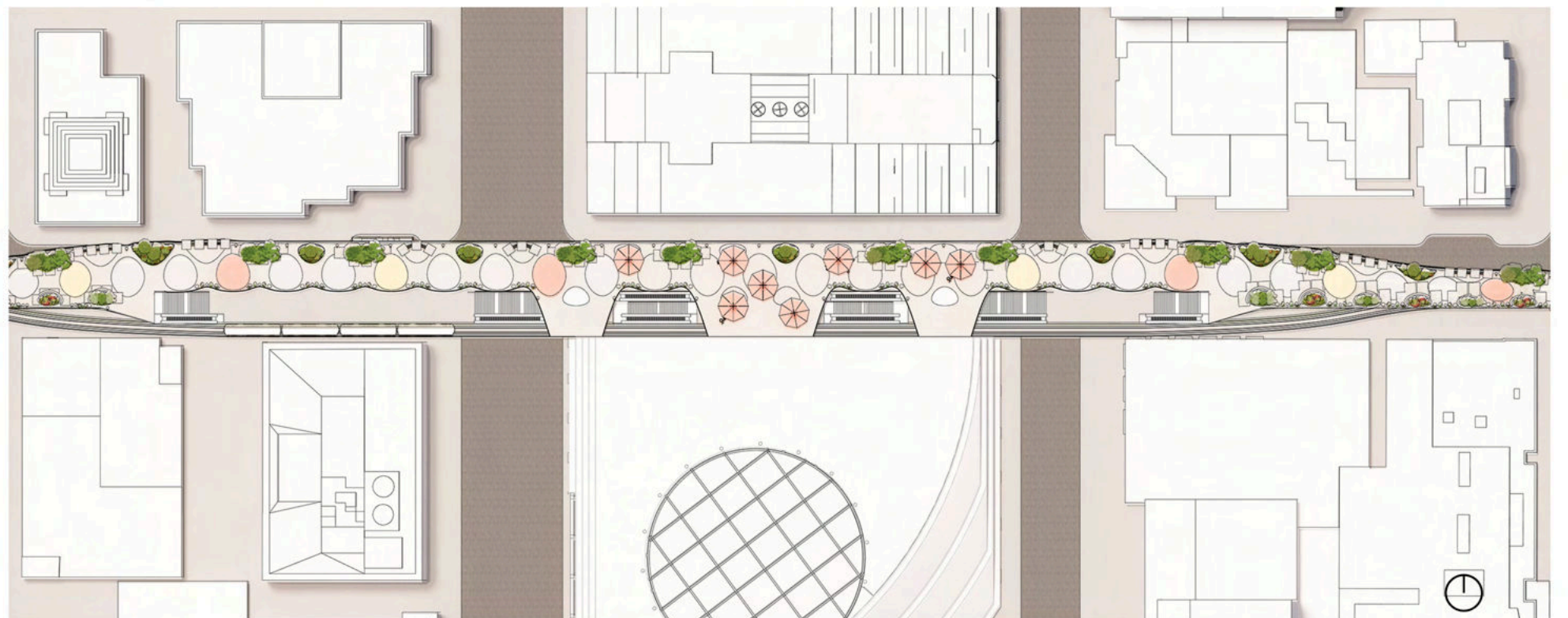
Existing Site Plan



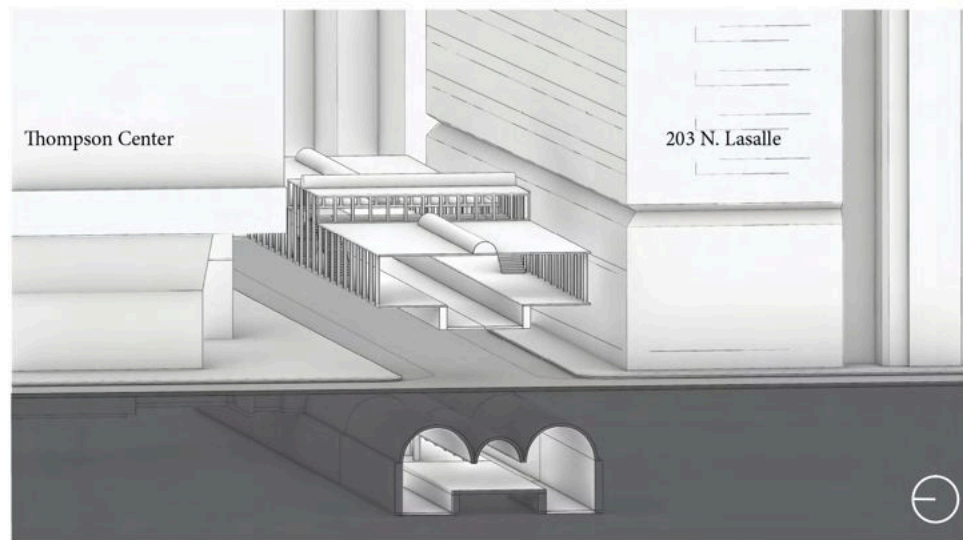
Existing L-Station Platform Level



Existing Ground Level



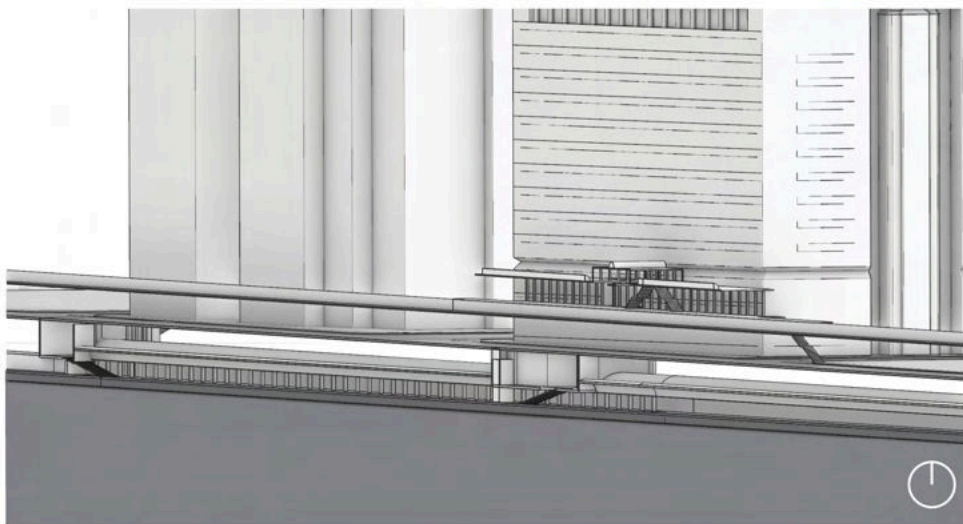
New Site Plan



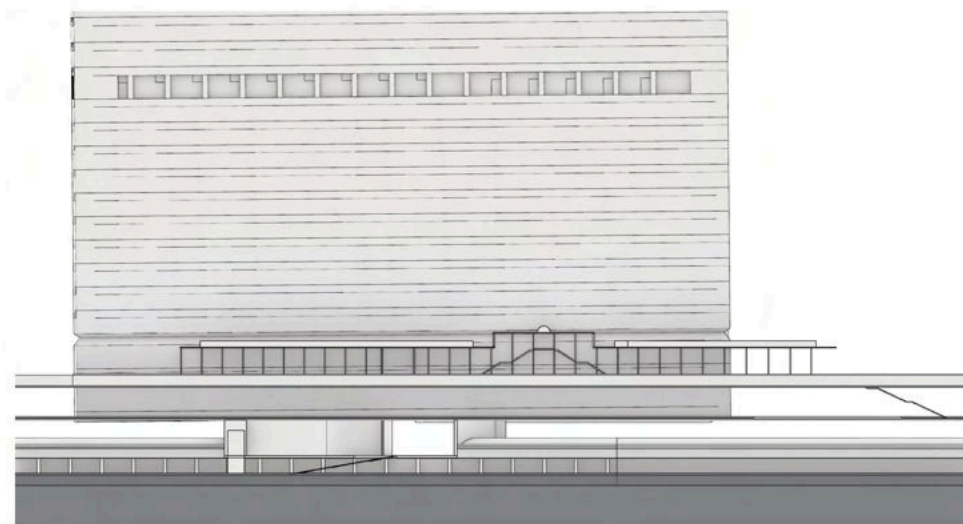
Section Perspective 1



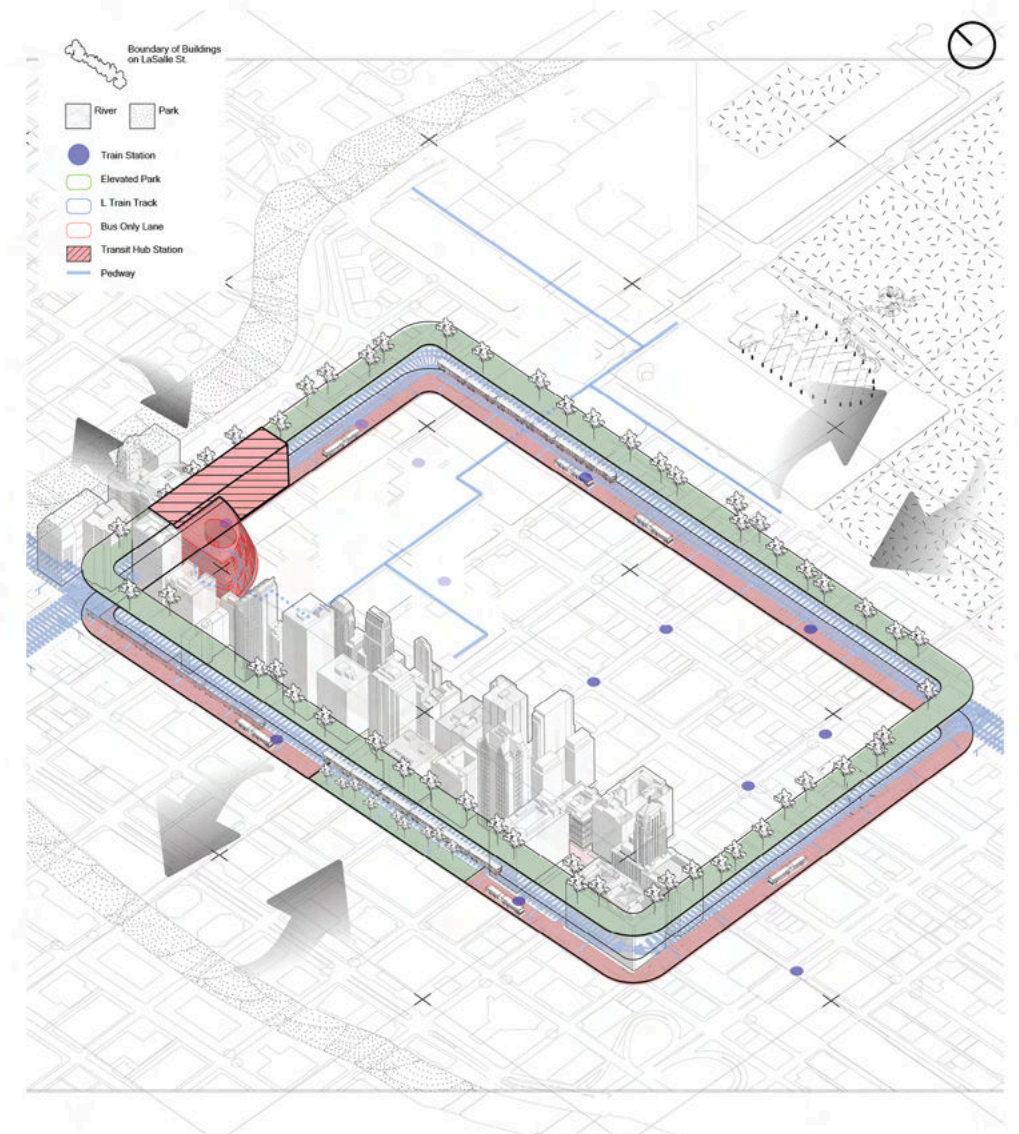
Section 1



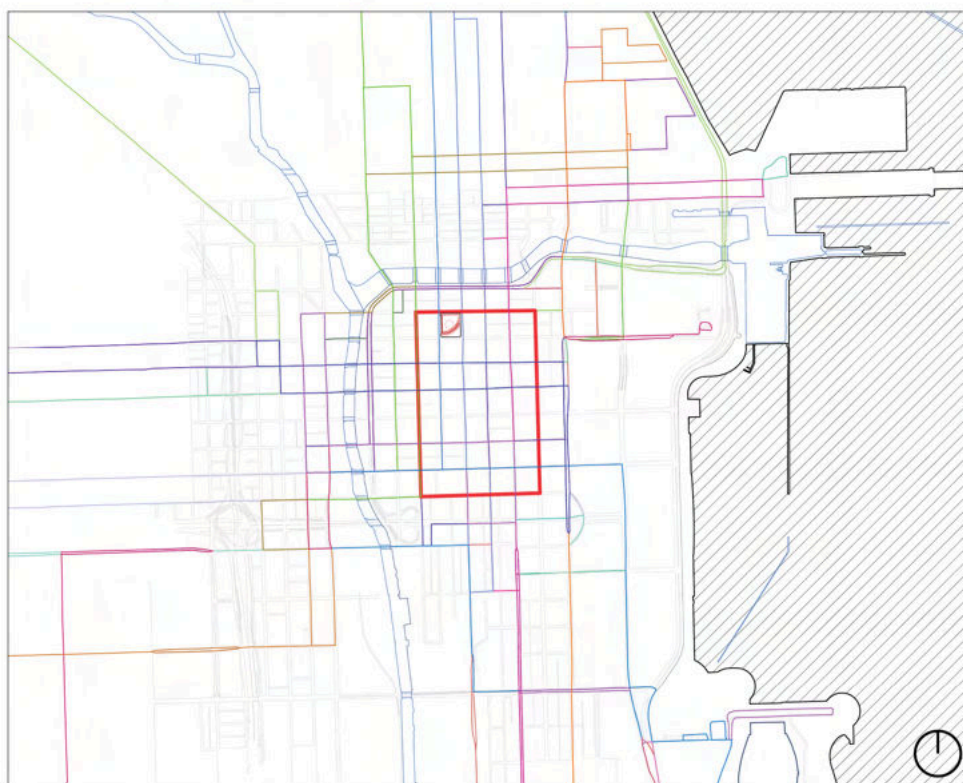
Section Perspective 2



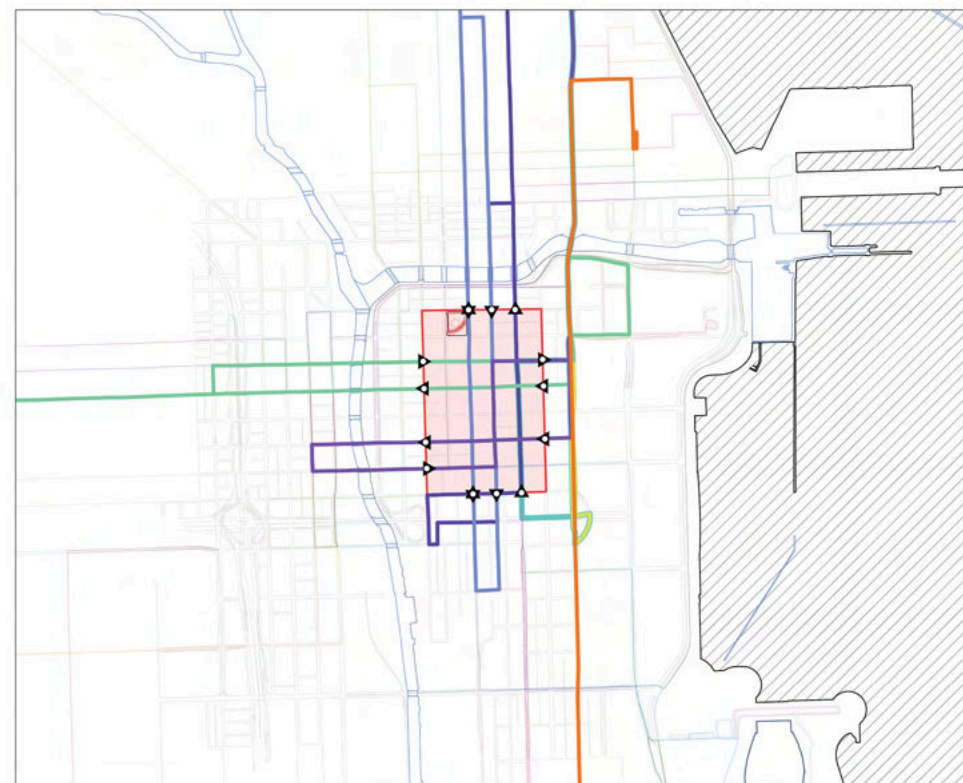
Section 2



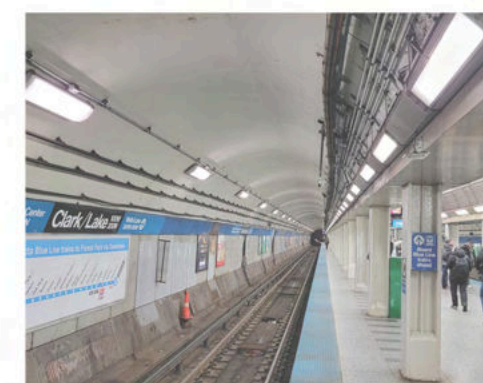
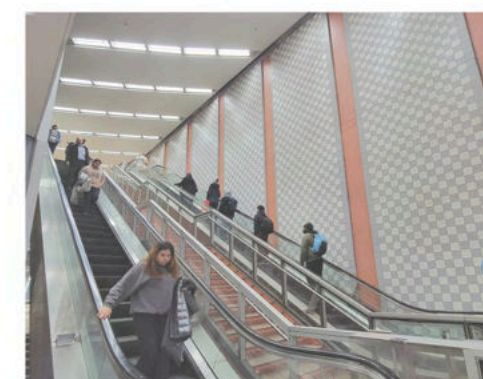
New Circulation



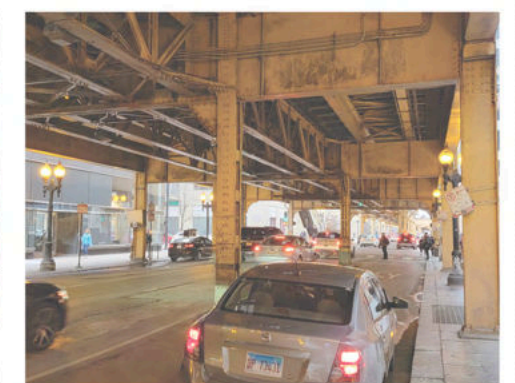
Existing Bus Routes

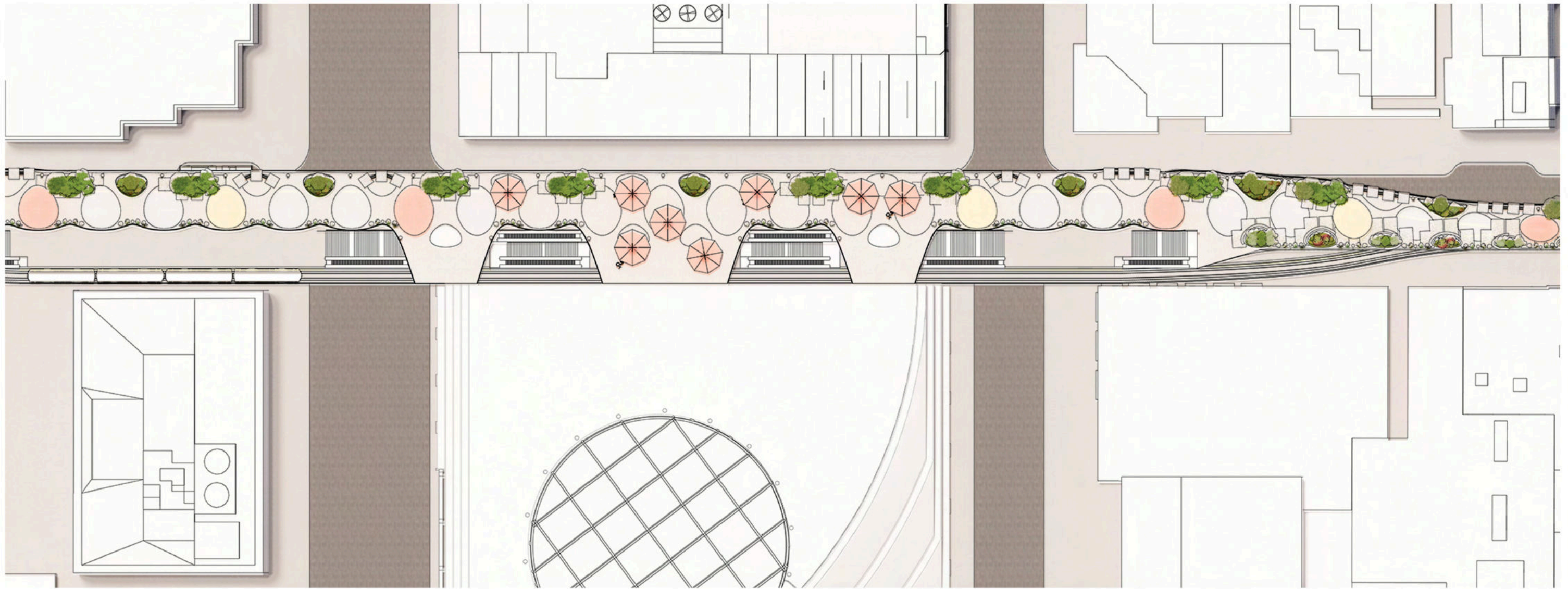


New Bus Routes



Existing Light Conditions





Longitudinal Section Perspective



Latitudinal Section



Ground Condition



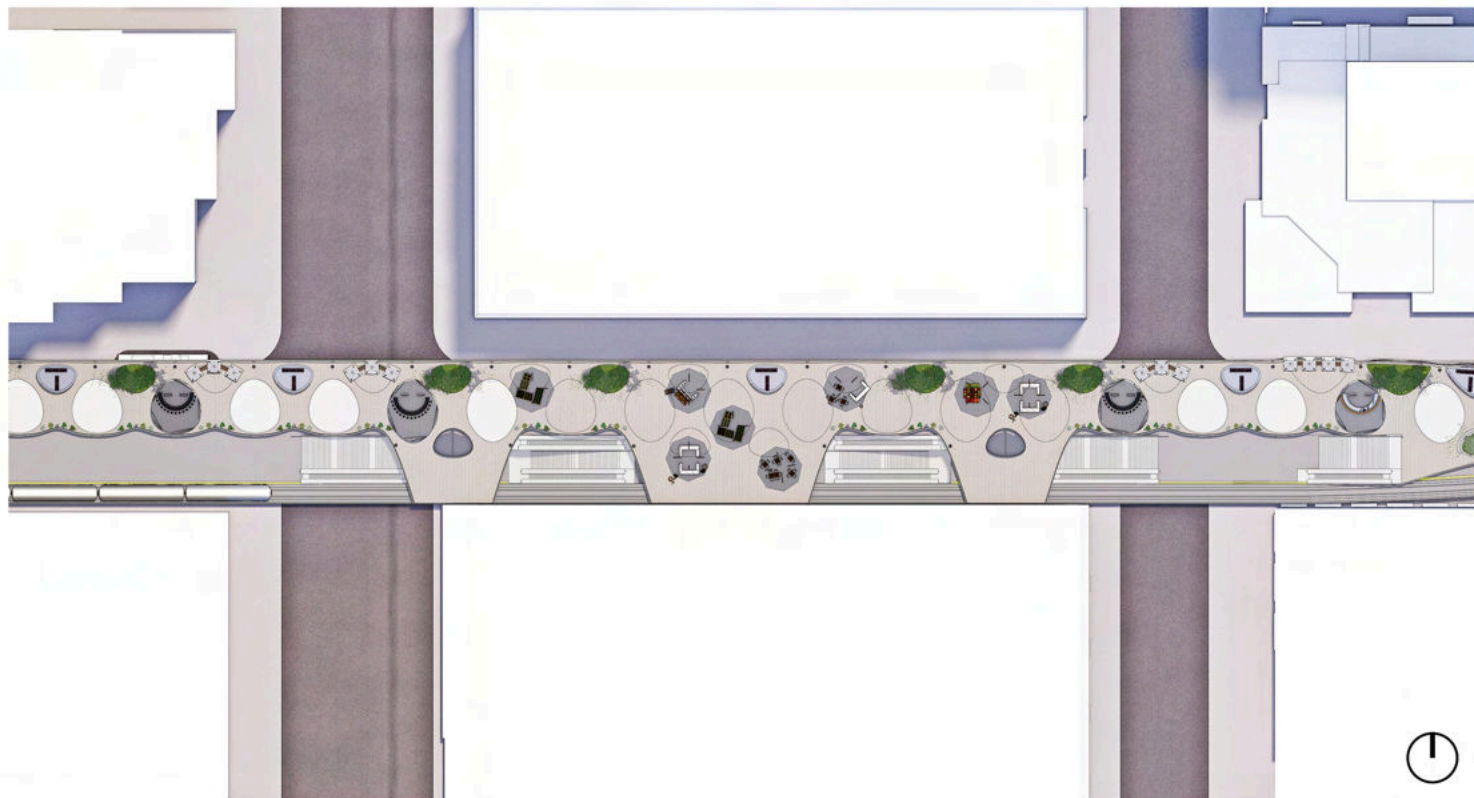
Bus Lane



Chi-Line



Grand Stair



Chi-Line Plan



CTA Train Line Plan

FISK GENERATING STATION

An Equitable Food Distribution Network

Year: *Fall 2021 - Spring 2022*

Program: *Adaptive Reuse - Food Commons*

Type: *Academic Work*

Location: *1111 W. Cermak Rd, Chicago, IL*

Teammate: *Julian Go*

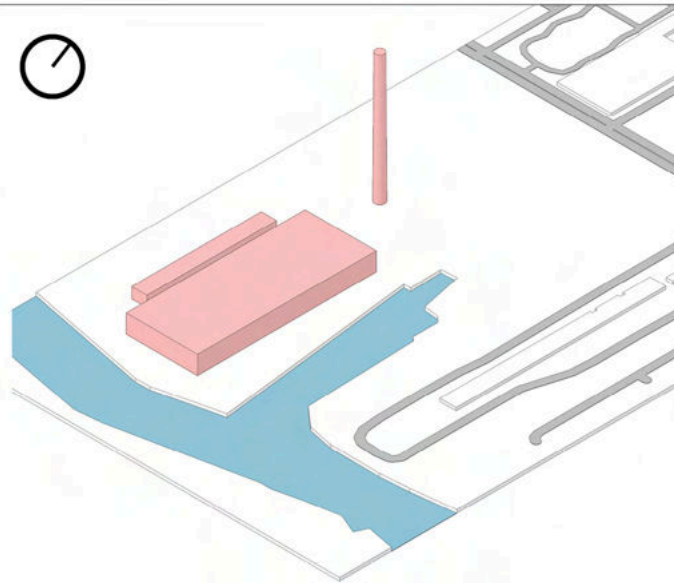
Tutor: *Andy Tinucci & Kindon Mills*

The Fisk Generating Station has had a significant impact on the West Loop. The factory went into service on October 2, 1903. It was built by the Chicago Edison Company and designed by Shepley, Rutan, and Coolidge. The original design featured two significant buildings constructed of load-bearing masonry walls with steel truss supports. In 1929, an additional Switch House was added to the site and was designed to mimic the previous two designs. The station, which once employed hundreds of community members, was shut down in 2012 due to a strike, several deaths, and one of the most significant carbon emissions in the city. Our goal is to preserve the original historic building through adaptive reuse to once again serve the community of Pilsen. Our network begins at the station but, over time, can start to spread across the community.

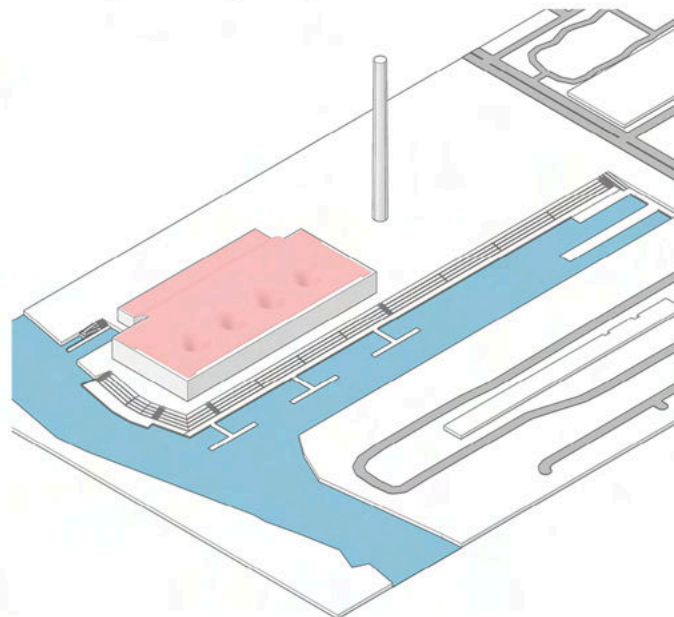


The now since abandoned Station stands as a reminder to the community of its History and Troubled past.

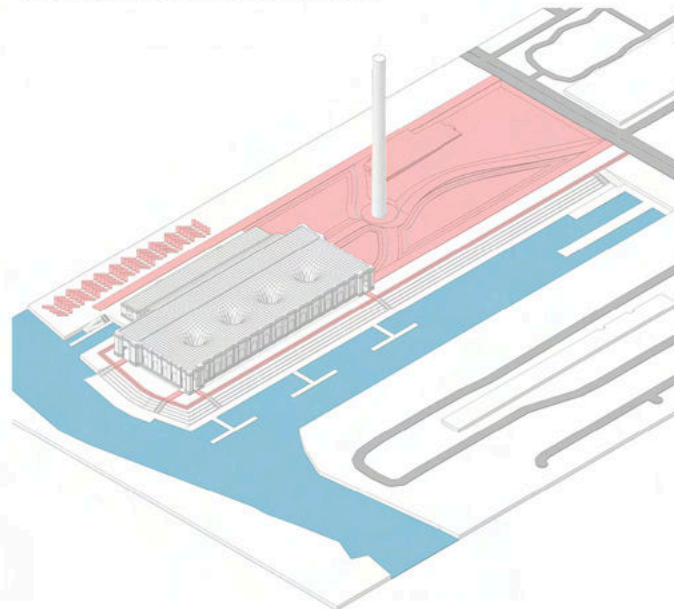




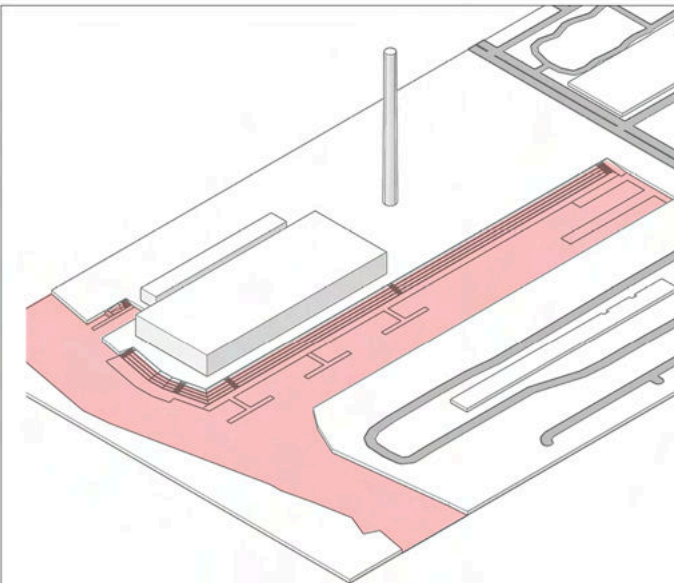
Existing Building Footprint



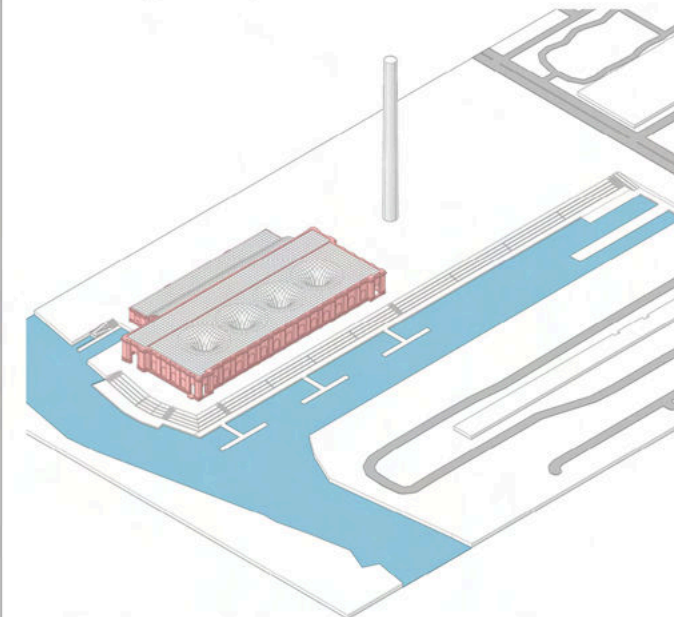
New Structural Intervention



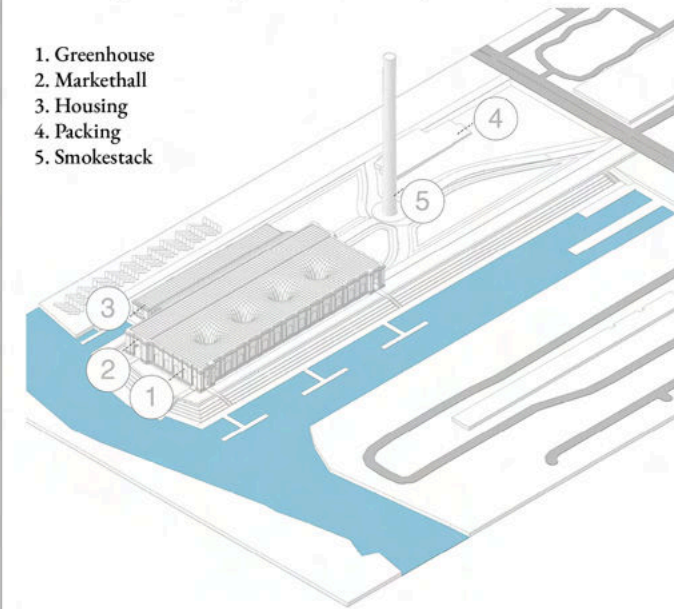
New Landscape Modifications



River Edge Changes

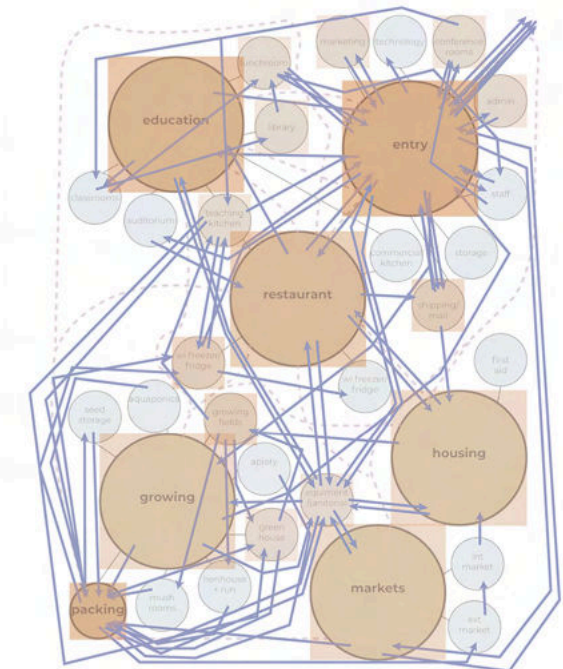


Existing Building Walls for Adaptive Reuse



Site Program

We followed the scar along the river to the lower West Loop, a primarily residential area. Nearly half of the existing infrastructure that needs to be in use is currently abandoned. Pilsen became the center of our exploration. With a rich Hispanic history, the neighborhood has quickly gentrified over the last several years as younger residents have begun to buy cheaper property and force previous generations out. Our building could serve as an urban food commons and a rehabilitation center. Our site was selected based on its accessibility from West Cermak and the river. In addition, it is within a walkable distance for both members of the community and local schools.



Combined Circulation Paths



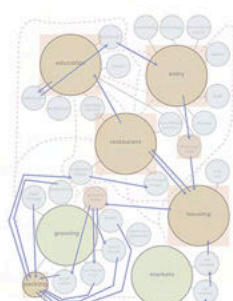
Community



Office Worker



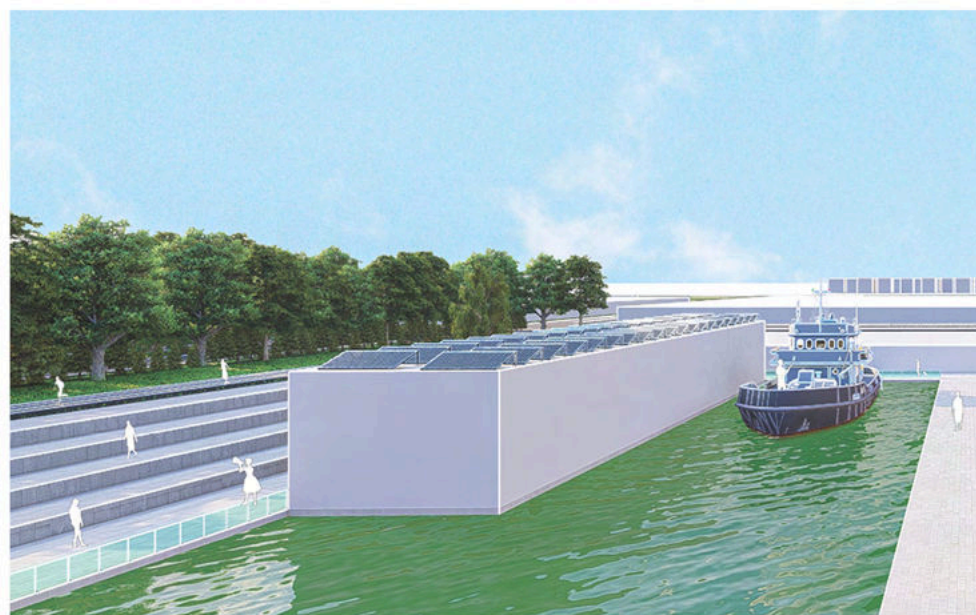
Teacher



Tenant



Inner Walkway with Vertical Farming



Rivertaxi Terminal



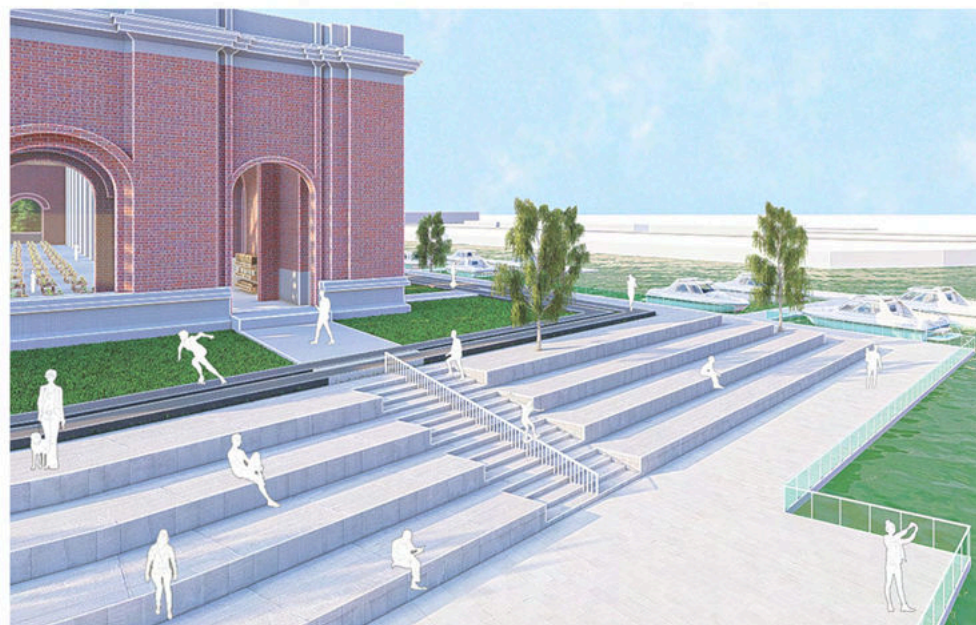
Wall Module Closed



Wall Module Open



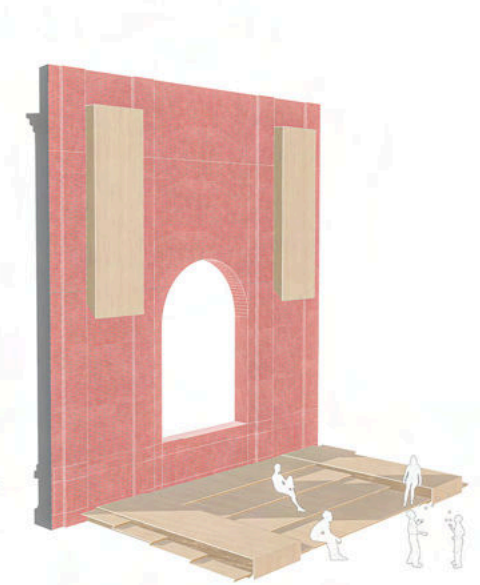
Bioswale, Water runoff strategy



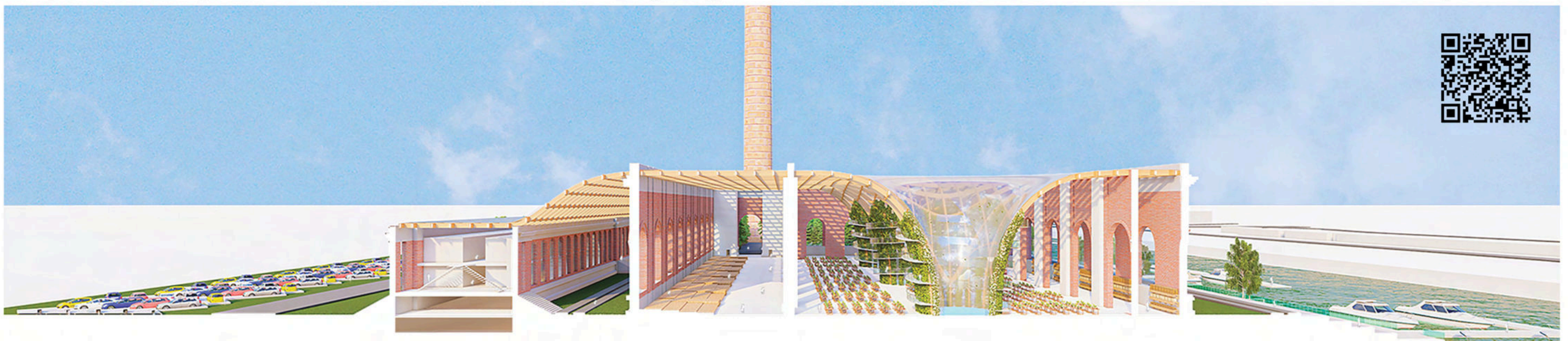
Riverwalk, Site Edge Strategy



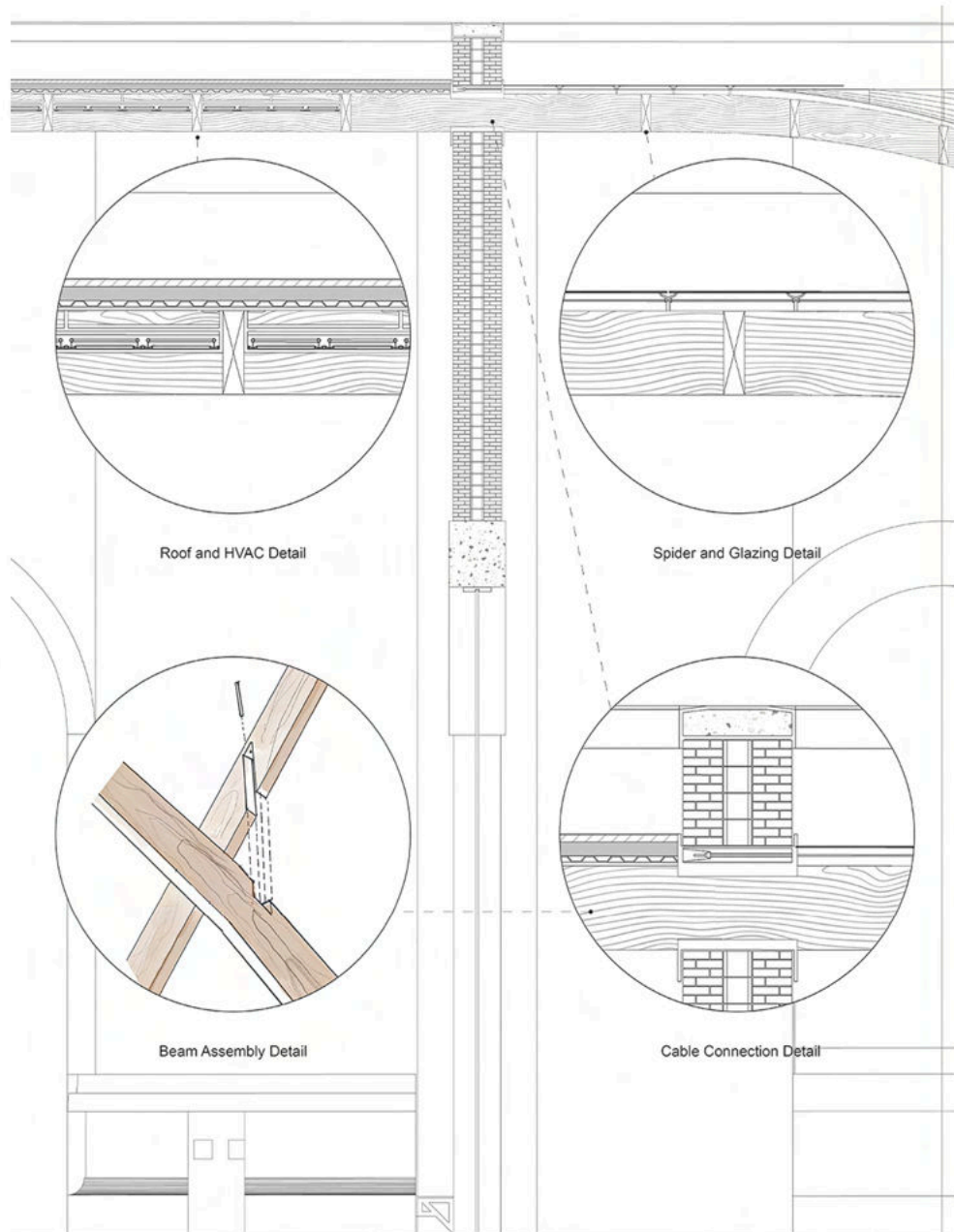
Wall Module Extended



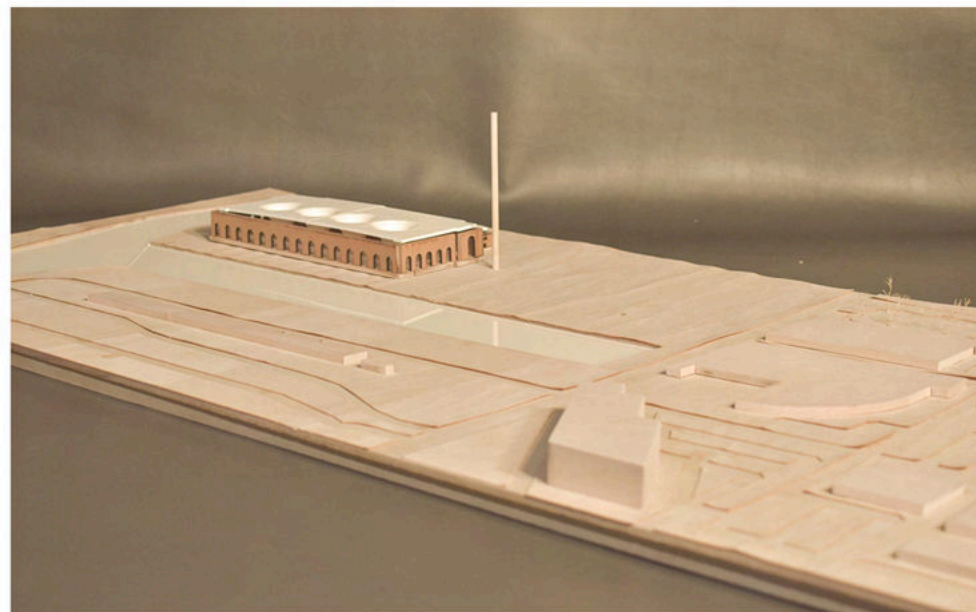
Wall Module Seating



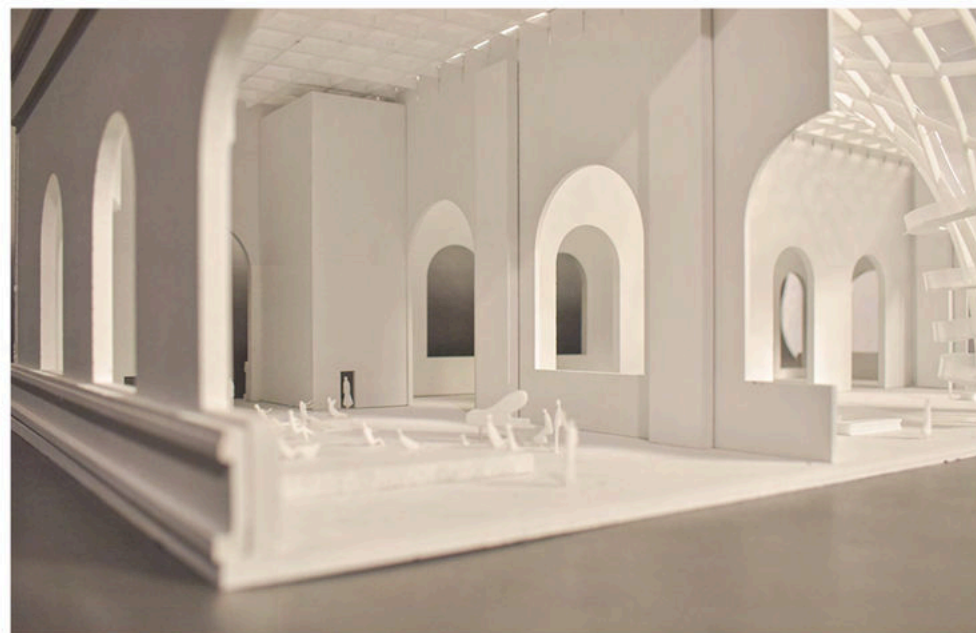
Section Perspective



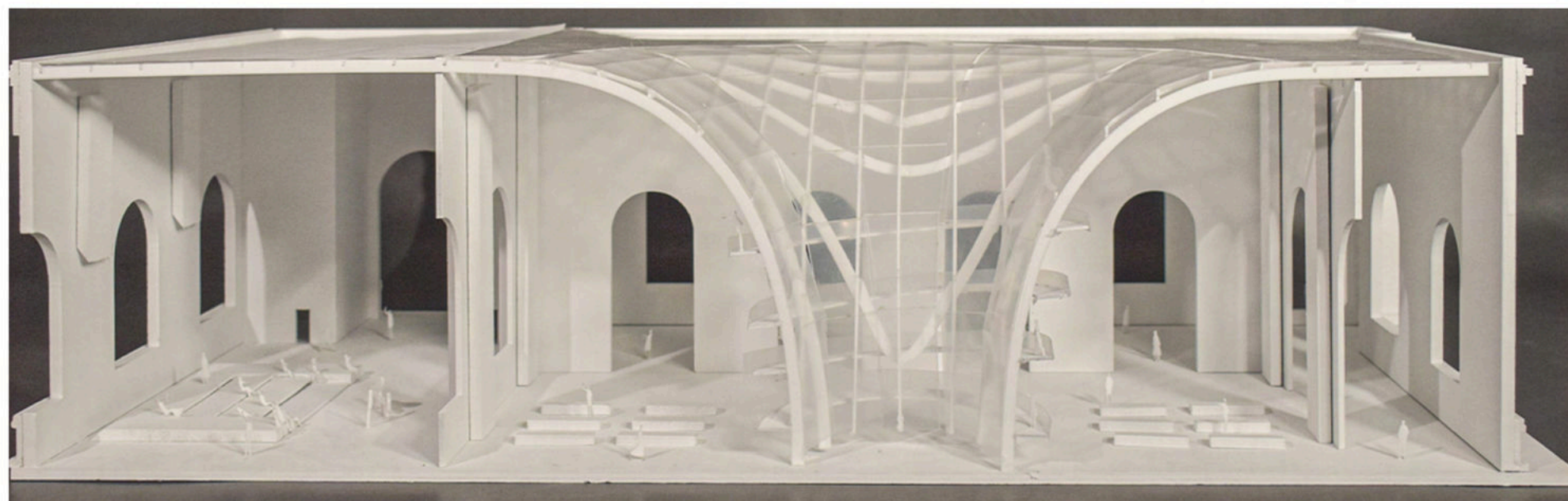
Wall Section Detail



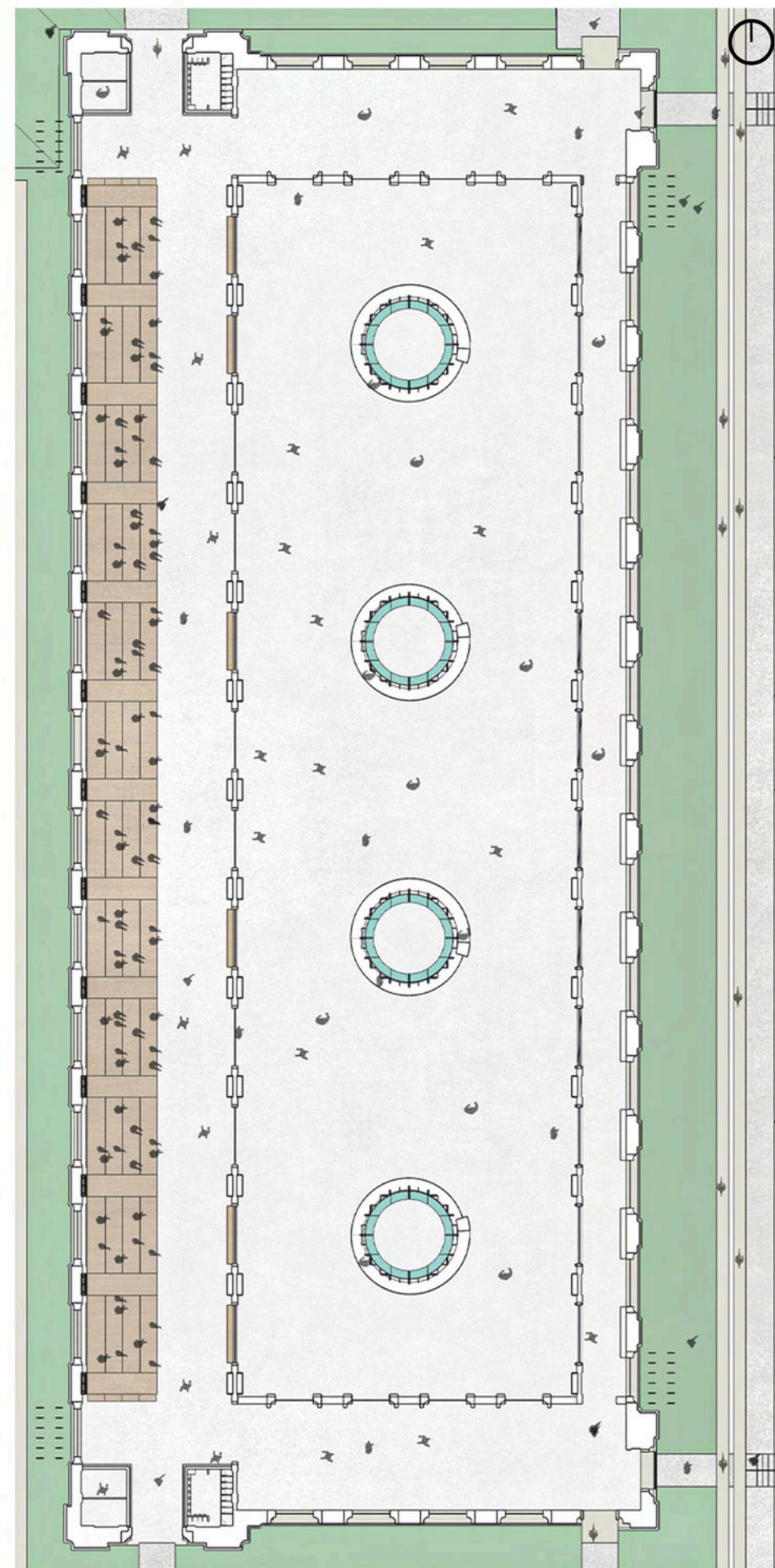
Overall Site Model



Model Interior



Model Section



Zoomed Plan with Seating Opened

CORE Projects

Professional

Year: *July 2025-Current*

Type: *Non-Academic Work*

Team: *Nick Mundell, Viktoria Henriksson, &
Danny Karas*

Firm: *Core studio at Thornton Tomasetti*

As a Computational Designer anchored in the global CORE Studio Modeling Group, my professional practice is defined by high-level technical collaboration with distributed teams across New York, Chicago, Philadelphia, London, Los Angeles, Mumbai, and Stockholm. I contribute to the realization of ambitious architectural typologies ranging from the aerodynamic verticality of the Jeddah Tower and 30 Hudson Yards to the complex, non-standard envelopes of the A's Las Vegas Ballpark and Qiddiya City Esports district. My technical workflow relies heavily on orchestrating seamless interoperability between geometric modeling and structural performance; I utilize Rhino.Inside.Revit and Grasshopper as primary drivers to synchronize data with analytical platforms like ETABS, SAP2000, and Tekla. This process is further augmented by the rigorous application of proprietary CORE Studio toolsets, enabling the rationalization of intricate forms that standard software cannot address. Beyond project delivery, I actively foster internal growth by developing training materials and leading bi-weekly "office hours" sessions to mentor colleagues on emerging workflows. This dedication to industry innovation extends to the wider community, where I have played a dual role in organizing logistics and competing in hackathons at both AECTech Boston 2025 and AECTech NYC 2025, consistently helping to shape the future of digital practice.



Jeddah Tower



Lincoln Center Damrosch Park



A's Las Vegas Ballpark



30 Hudson Yards



Hackathon Team at AECTech NYC 2025



Prince Mohammed bin Salman



Qiddiya City Esports

KPF Plugins

Professional

Year: *Summer 2024*

Type: *Non-Academic Work*

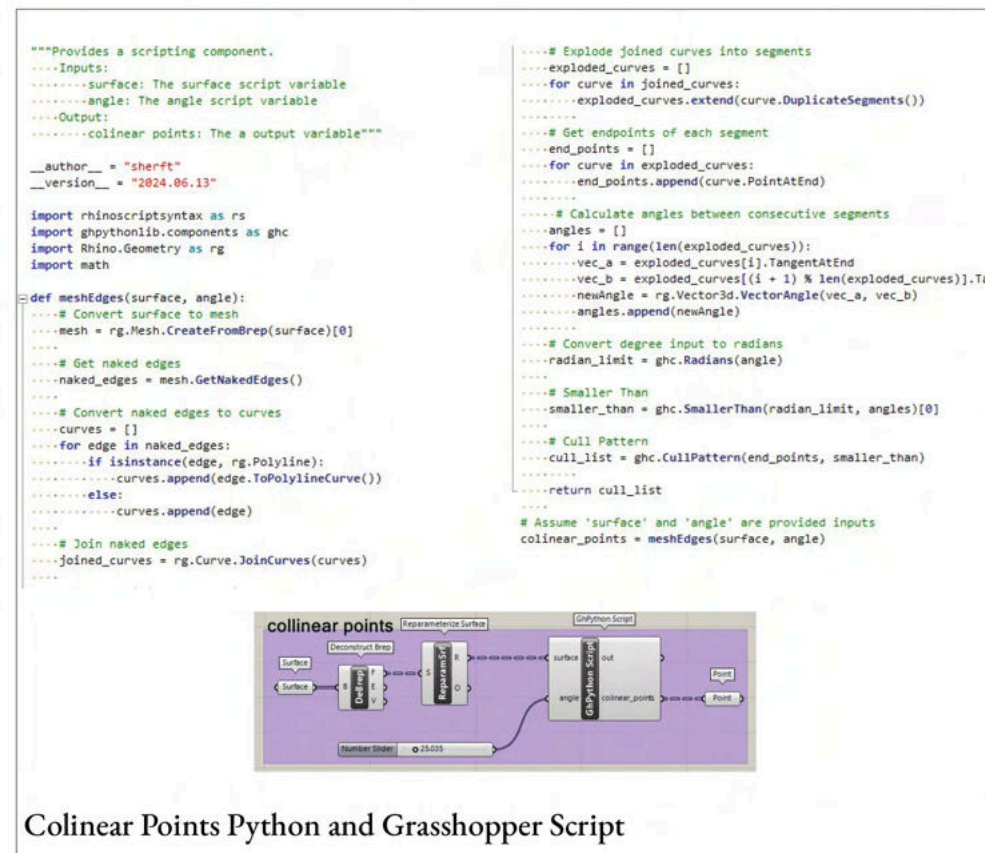
Team: *Joe Brennan, Madeleine Eggers &
Hazel Villena*

Firm: *Kohn Pederson Fox Associates PC*

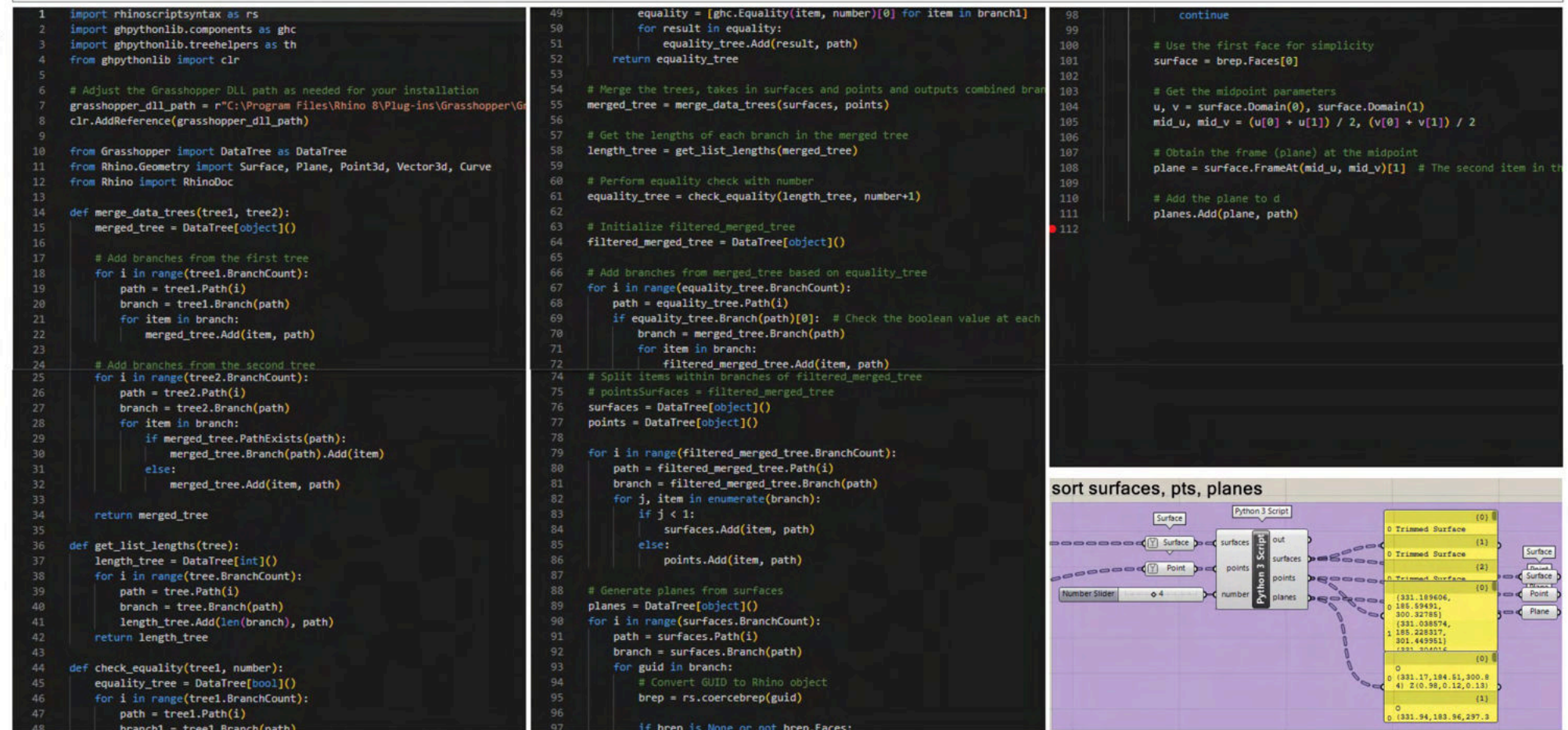
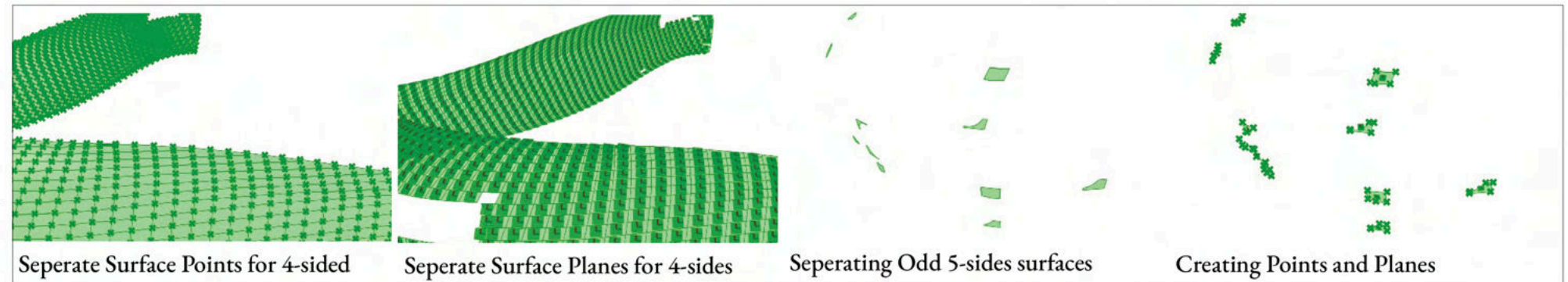
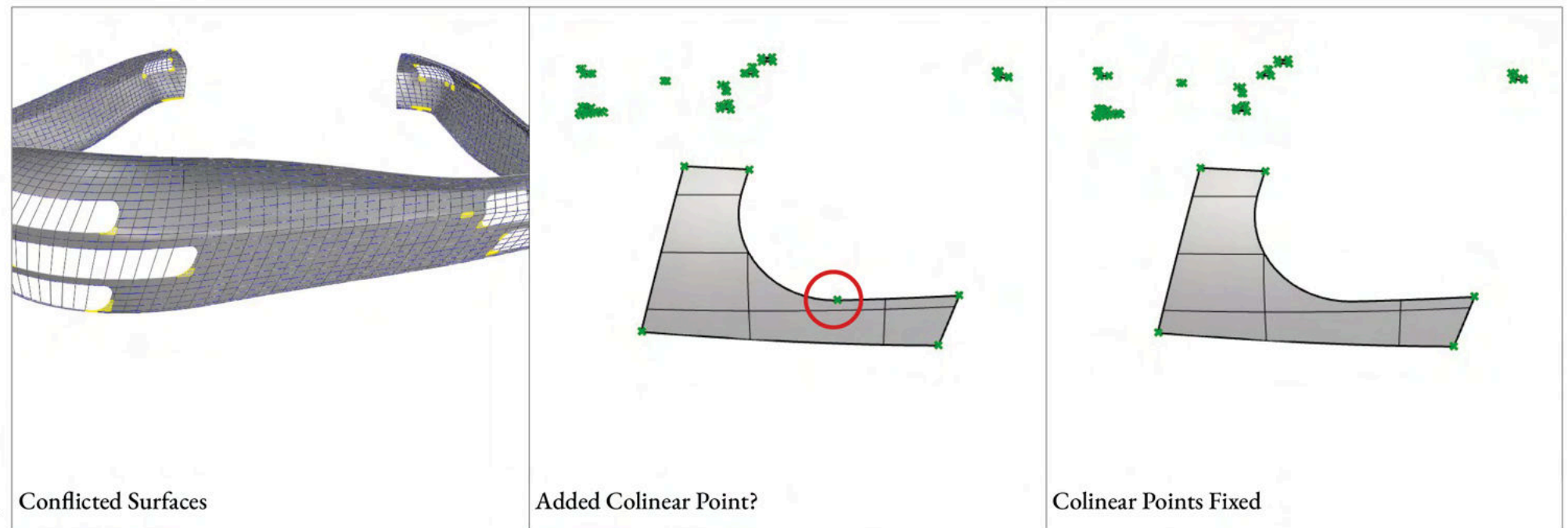
Link to Documentation: <https://bit.ly/4hb5bsK>

At KPF, I was part of the Computation Team, where I developed a suite of tools to address common issues encountered in project workflows. These tools were initially built using ghPython, then converted to C# to improve runtime performance and efficiency. Testing was conducted on a sample cloud project, which presented unique challenges, particularly with facade surface issues.

The plugins I created included several targeted functions. For instance, Sort Colinear Points corrected inaccuracies in surface point alignment by using a specified angle to improve accuracy. Another tool, Sort Sides, took a tree of surfaces and used Sort Colinear Points to arrange the surfaces by their colinear points. This plugin organized points and sides clockwise, positioning the top side as the first side and the top-left corner as the starting point for doubly curved surfaces. Additionally, I worked on Sort UV, a tool to align UVs of surfaces either to a base surface or along specified axes.



Colinear Points Python and Grasshopper Script



Sort Surfaces, Points, Planes Python and Grasshopper Script

POKE POKE & LOWCOUNTRY CHINATOWN

Professional

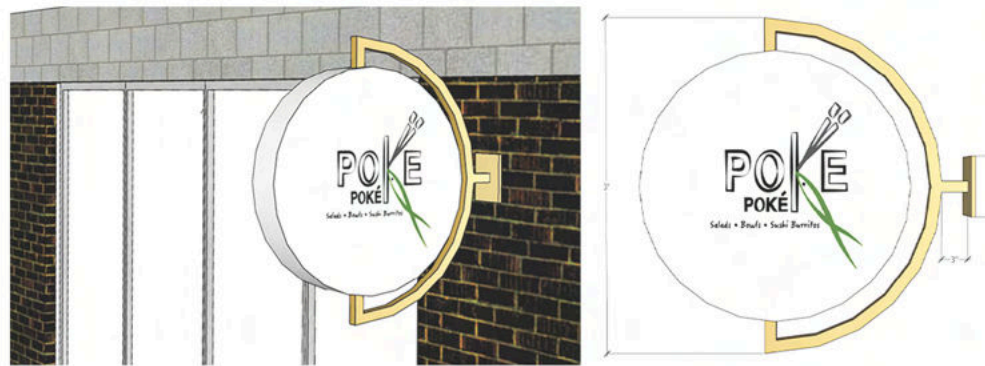
Year: *Summer 2021*

Type: *Non-Academic Work*

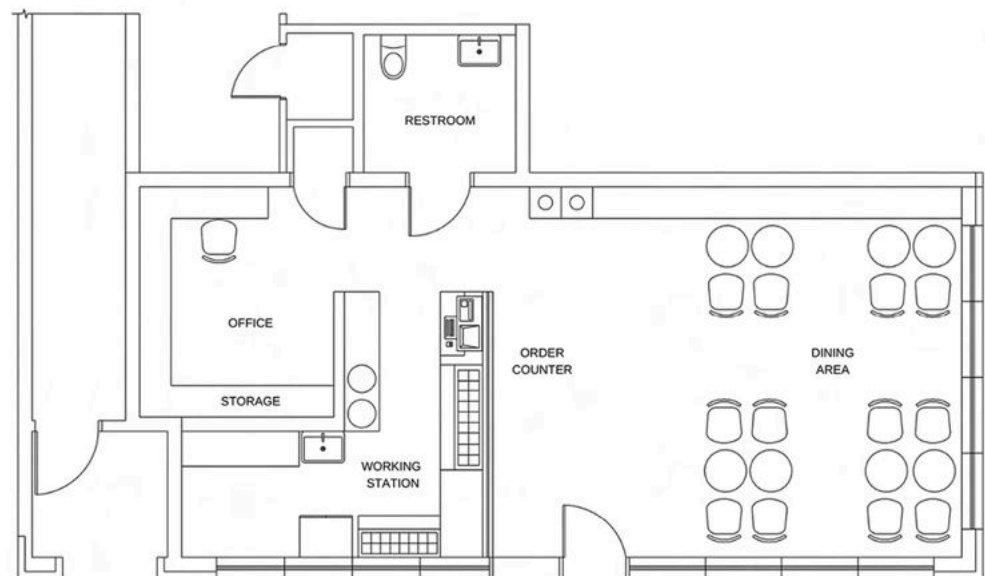
Team: *Kiery Calica, Paige Candelaria, Kee Chan, and Kinga Felker*

Firm: *Keyconcept*

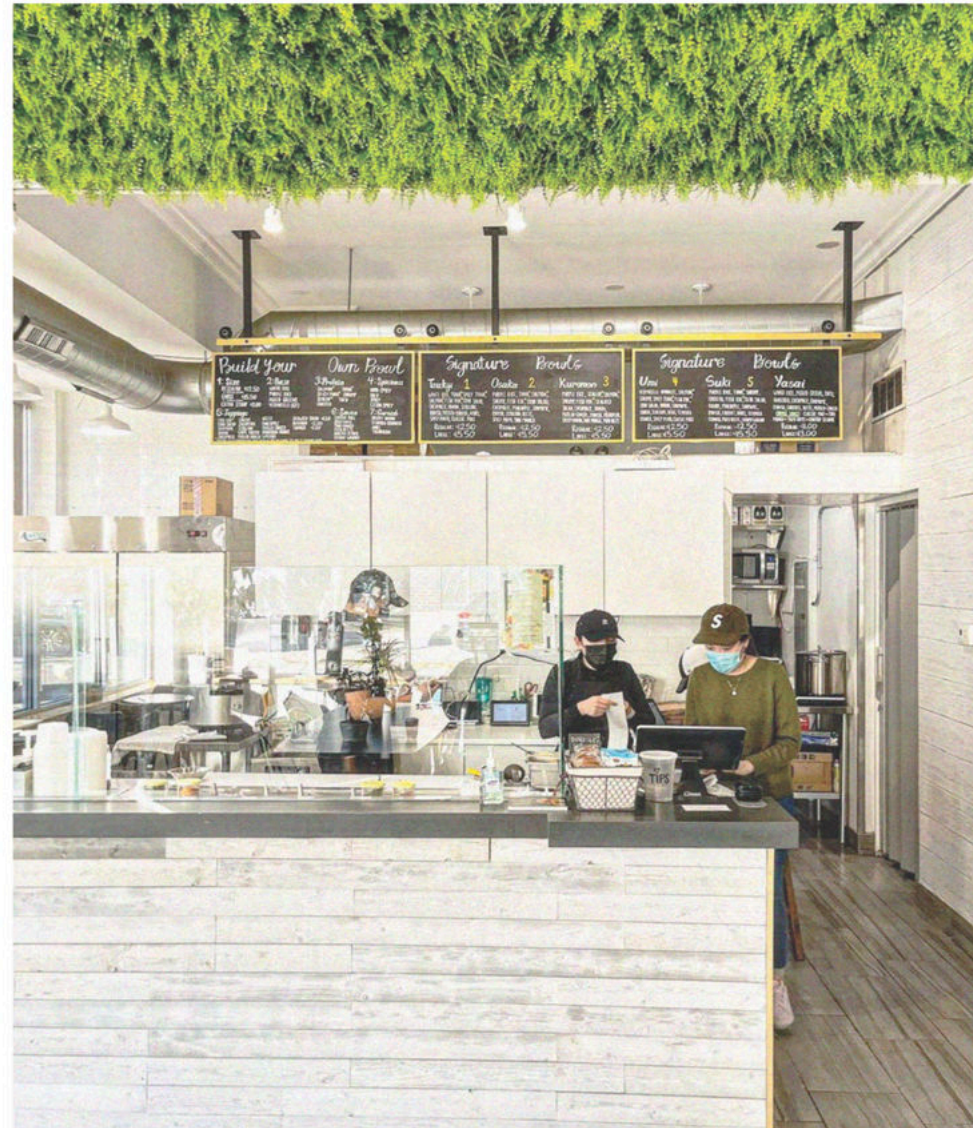
Poke Poke was a project that I worked on at KeyConcept. Key elements of the project included measuring and creating a 3D space of the pre-existing space, understanding what Poke Poke was and what the owner wanted in our representation of the area. I specifically worked on the dining space's plans, interior decorations, and wallpaper laminations.



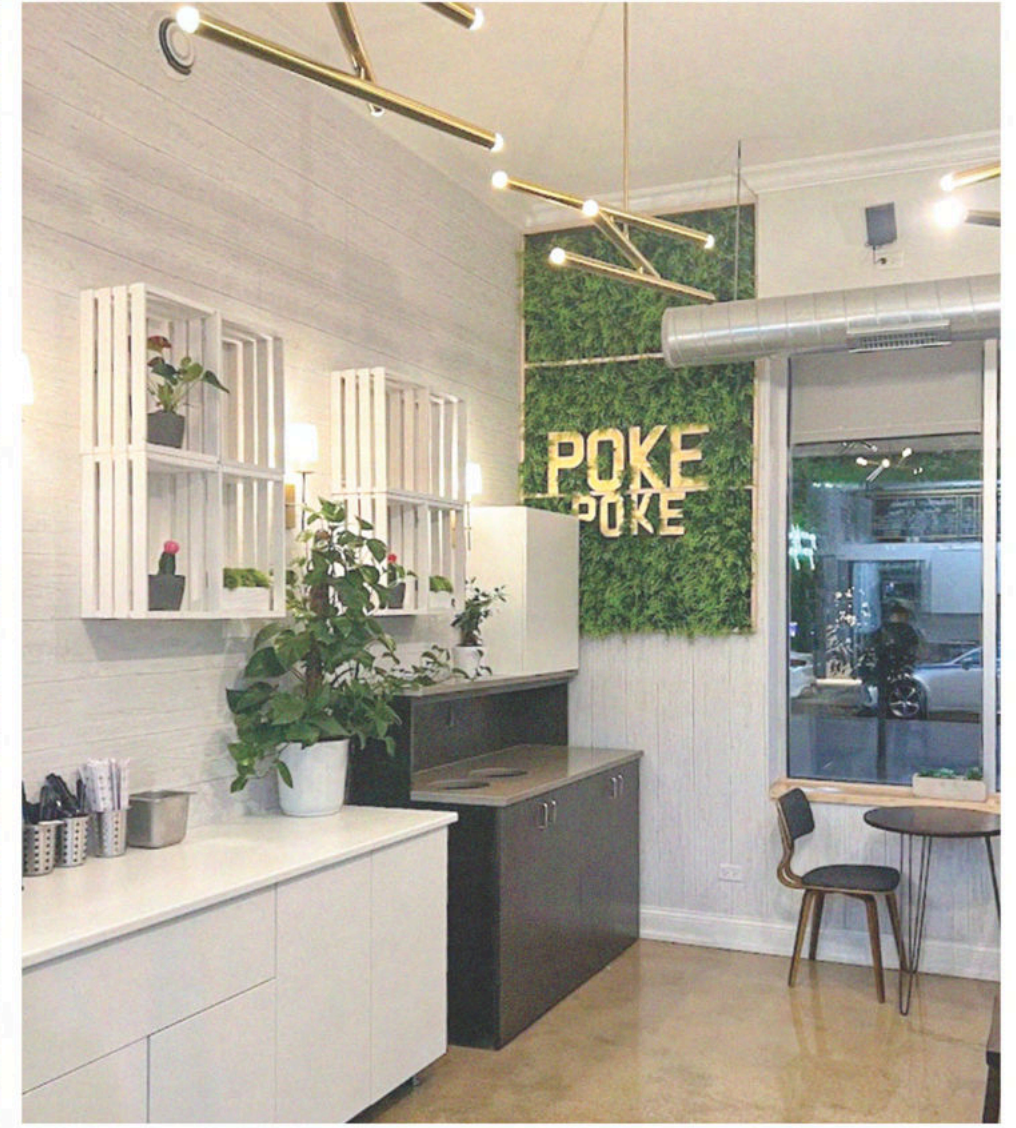
Poke Poke Signage



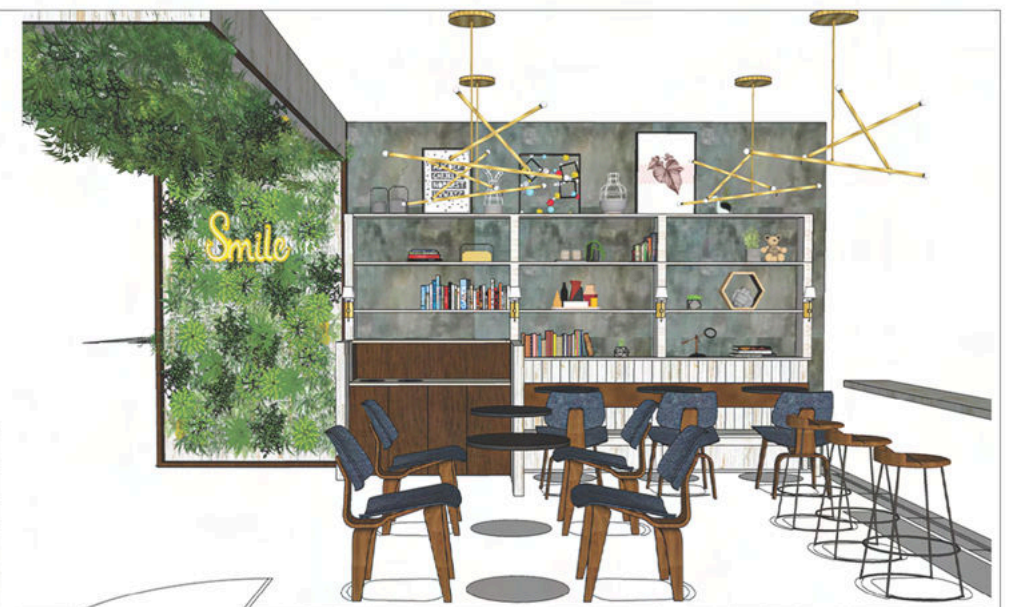
Interior Layout



Kitchen and Dining Space Completed Images



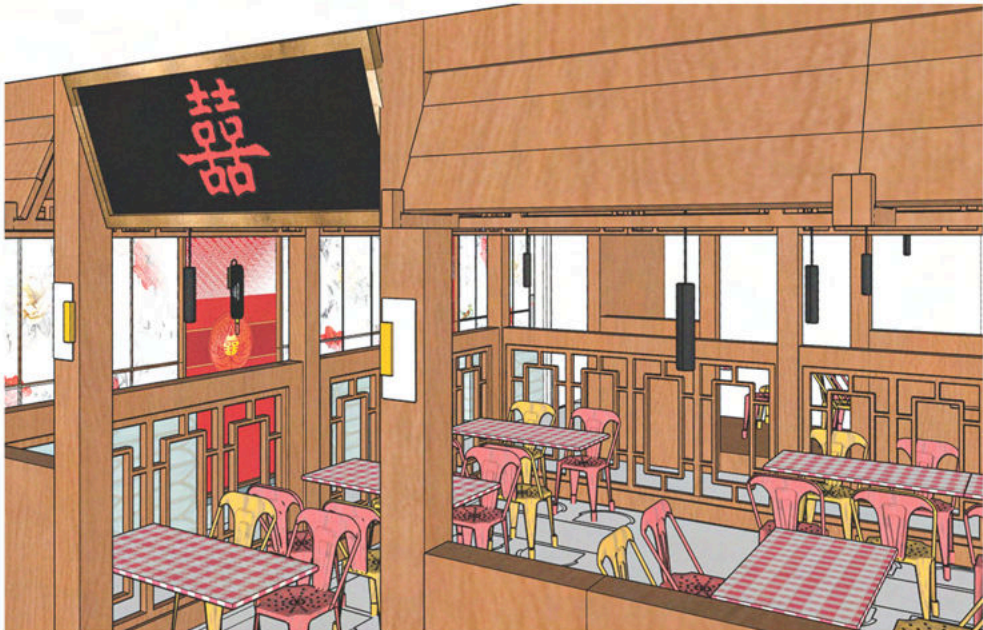
Kitchen and Dining 3D Models



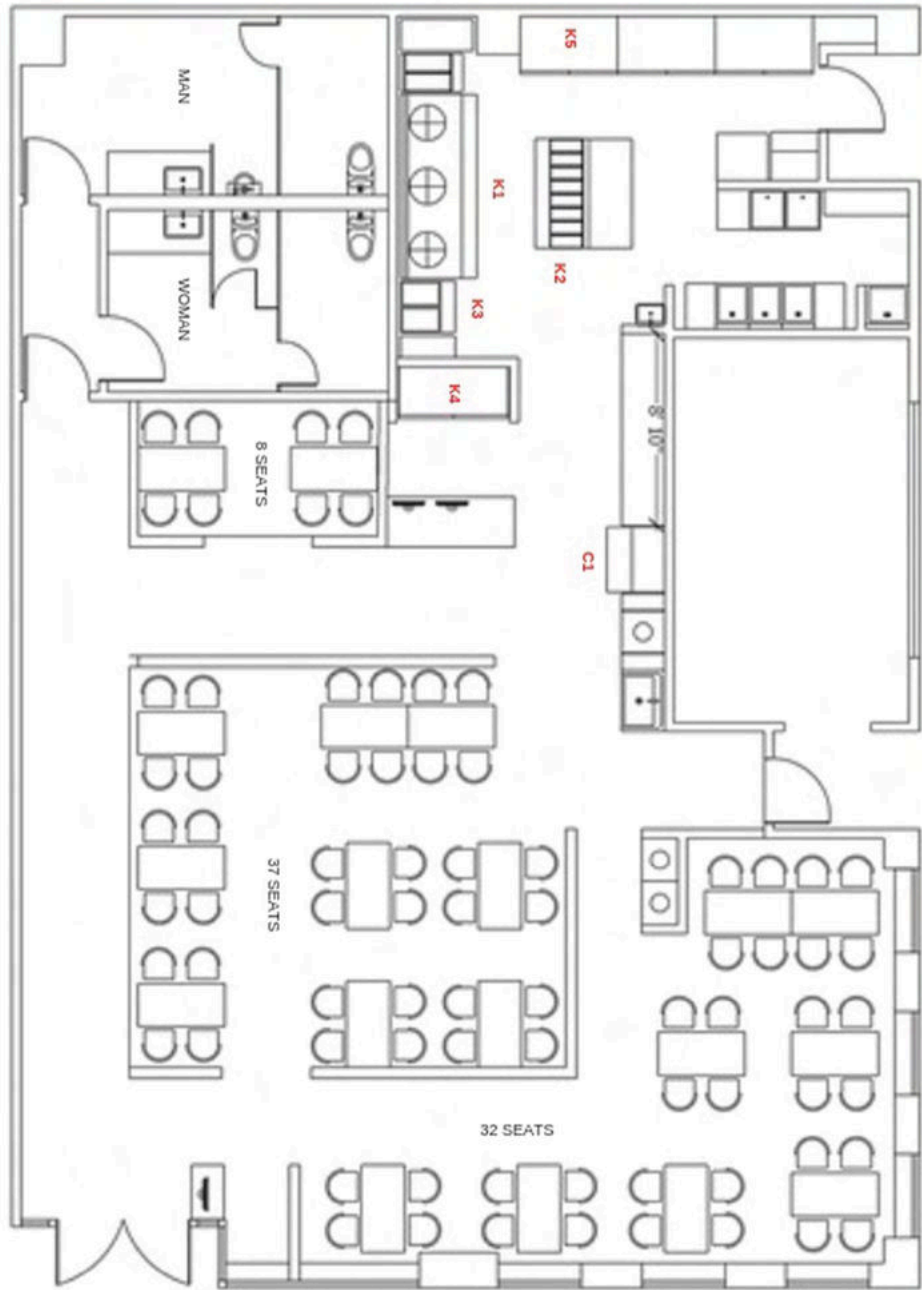
Our process for designing the space at Lowcountry Chinatown started by understanding the context from prior Lowcountry's designs in Chicago. By familiarizing ourselves with the space's interior, I produced iterations of my envisionment of the area, by mapping out the space in SketchUp to create a digital twin we could refer to during the construction phase.



Dining Space Completed Images



Entry and Dining Space 3D Models



Interior Layout

THANK YOU

Selected Works 2018-2026 | shandonherft@gmail.com

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